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# Google Colab Bengali Character Fast 14M Classifier
# Architecture:
# Fast LeNet-style stem -> residual CNN stages -> lightweight
token/MLP mixer -> classifier.
# Designed for very fast training with about 1.4 crore parameters.
#
# Expected zip layout:
# Dataset.zip
#   train/
#     class_1/
#       image001.bmp
#     class_2/
#       image002.bmp
#   test/
#     class_1/
#       image003.bmp
#     class_2/
#       image004.bmp
#
# In Colab:
# 1. Runtime > Change runtime type > GPU
# 2. Paste/run this file in one notebook cell.

import os
import random
import shutil
import time
import zipfile
from pathlib import Path
from typing import Dict, List, Optional, Tuple

import pandas as pd
import torch
import torch.nn as nn
import torch.optim as optim
from google.colab import files
from PIL import Image
from sklearn.metrics import classification_report, confusion_matrix
from torch.utils.data import DataLoader, Dataset
from torchvision import transforms
from torchvision.datasets import ImageFolder

SEED = 42
ZIP_LIMIT_MB = 500
DATA_ROOT = Path("/content/bengali_hybrid_dataset")
OUTPUT_DIR = Path("/content/bengali_hybrid_results")

# If upload button does not appear, upload zip from Colab's left Files
panel

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# and set ZIP_PATH = "/content/your_file.zip".
ZIP_PATH = ""
AUTO_USE_NEWEST_CONTENT_ZIP = True

MODEL_IMAGE_SIZE = 64
BATCH_SIZE = 256
EPOCHS = 20
LEARNING_RATE = 2e-3
WEIGHT_DECAY = 1e-4
NUM_WORKERS = 4
MAX_TRAIN_IMAGES = 40000
MAX_TEST_IMAGES = 10000
USE_TORCH_COMPILE = True

def seed_everything(seed: int) -> None:
    random.seed(seed)
    os.environ["PYTHONHASHSEED"] = str(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.benchmark = True

def choose_zip_file() -> Path:
    if ZIP_PATH:
        zip_path = Path(ZIP_PATH)
        if not zip_path.exists():
            raise FileNotFoundError(f"ZIP_PATH does not exist:
{zip_path}")
        return zip_path

    existing_zips = sorted(Path("/content").glob("*.zip"), key=lambda
p: p.stat().st_mtime, reverse=True)
    if AUTO_USE_NEWEST_CONTENT_ZIP and existing_zips:
        zip_path = existing_zips[0]
        print(f"Using existing zip from /content: {zip_path}")
        return zip_path

    print("Upload your train/test zip file now.")
    print("If the upload button does not appear, upload the zip from
Colab's left Files panel and rerun.")
    uploaded = files.upload()
    if not uploaded:
        raise RuntimeError("No zip uploaded. Set ZIP_PATH or upload a
zip to /content and rerun.")
    zip_name = next(iter(uploaded.keys()))
    return Path("/content") / zip_name

def upload_and_extract_zip() -> Path:

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zip_path = choose_zip_file()
if zip_path.suffix.lower() != ".zip":
    raise ValueError("Please upload a .zip file.")

size_mb = zip_path.stat().st_size / (1024 * 1024)
if size_mb > ZIP_LIMIT_MB:
    raise ValueError(f"Zip is {size_mb:.1f} MB. Limit is
{ZIP_LIMIT_MB} MB.")

if DATA_ROOT.exists():
    shutil.rmtree(DATA_ROOT)
DATA_ROOT.mkdir(parents=True, exist_ok=True)

with zipfile.ZipFile(zip_path) as archive:
    for member in archive.infolist():
        target = DATA_ROOT / member.filename
        if not
str(target.resolve()).startswith(str(DATA_ROOT.resolve())):
            raise ValueError(f"Unsafe path inside zip:
{member.filename}")
        archive.extractall(DATA_ROOT)

print(f"Extracted to: {DATA_ROOT}")
return DATA_ROOT

def find_split_dir(root: Path, split_name: str) -> Path:
    direct = root / split_name
    if direct.exists() and direct.is_dir():
        return direct

    candidates = [p for p in root.rglob("*") if p.is_dir() and
p.name.lower() == split_name]
    if not candidates:
        raise FileNotFoundError(f"Could not find a '{split_name}'
folder inside the zip.")
    return candidates[0]

def count_images(split_dir: Path) -> int:
    extensions = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}
    return sum(1 for p in split_dir.rglob("*") if p.is_file() and
p.suffix.lower() in extensions)

class LimitedImageFolder(Dataset):
    def __init__(self, root: Path, transform, max_images:
Optional[int] = None):
        self.base = ImageFolder(str(root), transform=None)
        self.classes = self.base.classes

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self.class_to_idx = self.base.class_to_idx
self.transform = transform

samples = list(self.base.samples)
if max_images and len(samples) > max_images:
    by_class: Dict[int, List[Tuple[str, int]]] = {}
    for sample in samples:
        by_class.setdefault(sample[1], []).append(sample)
    limited = []
    per_class = max(1, max_images // max(1, len(by_class)))
    rng = random.Random(SEED)
    for class_samples in by_class.values():
        rng.shuffle(class_samples)
        limited.extend(class_samples[:per_class])
    samples = limited[:max_images]

self.samples = samples
self.targets = [target for _, target in samples]

def __len__(self):
    return len(self.samples)

def __getitem__(self, index):
    path, target = self.samples[index]
    image = Image.open(path).convert("RGB")
    if self.transform:
        image = self.transform(image)
    return image, target

class LeNetStem(nn.Module):
    def __init__(self):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=5, padding=2),
            nn.BatchNorm2d(32),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),
            nn.Conv2d(32, 64, kernel_size=5, padding=2),
            nn.BatchNorm2d(64),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(inplace=True),
        )

    def forward(self, x):
        return self.features(x)

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class ConvBNAct(nn.Module):
    def __init__(self, in_channels: int, out_channels: int, stride:
int = 1):
        super().__init__()
        self.block = nn.Sequential(
            nn.Conv2d(in_channels, out_channels, kernel_size=3,
stride=stride, padding=1, bias=False),
            nn.BatchNorm2d(out_channels),
            nn.SiLU(inplace=True),
        )

    def forward(self, x):
        return self.block(x)

class FastResidualBlock(nn.Module):
    def __init__(self, in_channels: int, out_channels: int, stride:
int = 1):
        super().__init__()
        self.conv1 = ConvBNAct(in_channels, out_channels,
stride=stride)
        self.conv2 = nn.Sequential(
            nn.Conv2d(out_channels, out_channels, kernel_size=3,
padding=1, bias=False),
            nn.BatchNorm2d(out_channels),
        )
        self.skip = (
            nn.Identity()
            if in_channels == out_channels and stride == 1
            else nn.Sequential(
                nn.Conv2d(in_channels, out_channels, kernel_size=1,
stride=stride, bias=False),
                nn.BatchNorm2d(out_channels),
            )
        )
        self.act = nn.SiLU(inplace=True)

    def forward(self, x):
        return self.act(self.conv2(self.conv1(x)) + self.skip(x))

class Fast14MCharacterNet(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.stem = nn.Sequential(
            ConvBNAct(3, 64),
            FastResidualBlock(64, 64),
            FastResidualBlock(64, 64),
        )

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        self.stage2 = nn.Sequential(
            FastResidualBlock(64, 128, stride=2),
            FastResidualBlock(128, 128),
        )
        self.stage3 = nn.Sequential(
            FastResidualBlock(128, 256, stride=2),
            FastResidualBlock(256, 256),
            FastResidualBlock(256, 256),
        )
        self.stage4 = nn.Sequential(
            FastResidualBlock(256, 512, stride=2),
            FastResidualBlock(512, 512),
        )
        self.pool = nn.AdaptiveAvgPool2d(1)
        self.mixer = nn.Sequential(
            nn.Linear(512, 2048),
            nn.SiLU(inplace=True),
            nn.Dropout(0.20),
            nn.Linear(2048, 512),
            nn.SiLU(inplace=True),
        )
        self.classifier = nn.Linear(512, num_classes)

    def forward(self, x):
        x = self.stem(x)
        x = self.stage2(x)
        x = self.stage3(x)
        x = self.stage4(x)
        x = self.pool(x).flatten(1)
        x = self.mixer(x)
        return self.classifier(x)

def get_transforms() -> Tuple[transforms.Compose, transforms.Compose]:
    train_tfms = transforms.Compose(
        [
            transforms.Resize((MODEL_IMAGE_SIZE, MODEL_IMAGE_SIZE)),
            transforms.RandomAffine(degrees=8, translate=(0.05, 0.05),
scale=(0.95, 1.05)),
            transforms.ToTensor(),
            transforms.Normalize(mean=(0.485, 0.456, 0.406),
std=(0.229, 0.224, 0.225)),
        ]
    )
    test_tfms = transforms.Compose(
        [
            transforms.Resize((MODEL_IMAGE_SIZE, MODEL_IMAGE_SIZE)),
            transforms.ToTensor(),
            transforms.Normalize(mean=(0.485, 0.456, 0.406),
std=(0.229, 0.224, 0.225)),
        ]
    )

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    ]
)
return train_tfms, test_tfms

def train_one_epoch(model, loader, criterion, optimizer, scheduler,
                    scaler, device) -> Tuple[float, float]:
    model.train()
    total_loss = 0.0
    correct = 0
    total = 0

    for images, labels in loader:
        images = images.to(device, non_blocking=True,
memory_format=torch.channels_last)
        labels = labels.to(device, non_blocking=True)

        optimizer.zero_grad(set_to_none=True)
        with torch.amp.autocast(device_type=device.type,
enabled=device.type == "cuda"):
            logits = model(images)
            loss = criterion(logits, labels)

            scaler.scale(loss).backward()
            scaler.step(optimizer)
            scaler.update()
            scheduler.step()

            total_loss += loss.item() * labels.size(0)
            correct += (logits.argmax(dim=1) == labels).sum().item()
            total += labels.size(0)

    return total_loss / max(1, total), correct / max(1, total)

@torch.no_grad()
def evaluate(model, loader, criterion, device) -> Tuple[float, float,
List[int], List[int]]:
    model.eval()
    total_loss = 0.0
    correct = 0
    total = 0
    y_true = []
    y_pred = []

    for images, labels in loader:
        images = images.to(device, non_blocking=True,
memory_format=torch.channels_last)
        labels = labels.to(device, non_blocking=True)
        logits = model(images)

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        loss = criterion(logits, labels)
        preds = logits.argmax(dim=1)

        total_loss += loss.item() * labels.size(0)
        correct += (preds == labels).sum().item()
        total += labels.size(0)
        y_true.extend(labels.cpu().tolist())
        y_pred.extend(preds.cpu().tolist())

    return total_loss / max(1, total), correct / max(1, total),
    y_true, y_pred

def fit_model(num_classes: int, train_loader, test_loader, device):
    model = Fast14MCharacterNet(num_classes).to(device,
memory_format=torch.channels_last)
    if USE_TORCH_COMPILE and hasattr(torch, "compile"):
        try:
            model = torch.compile(model)
            print("torch.compile enabled")
        except Exception as exc:
            print(f"torch.compile skipped: {exc}")

    trainable_params = [p for p in model.parameters() if
p.requires_grad]
    param_count = sum(p.numel() for p in trainable_params)
    print(f"Trainable parameters: {param_count:,} ({param_count /
10_000_000:.2f} crore)")

    try:
        optimizer = optim.AdamW(trainable_params, lr=LEARNING_RATE,
weight_decay=WEIGHT_DECAY, fused=device.type == "cuda")
    except TypeError:
        optimizer = optim.AdamW(trainable_params, lr=LEARNING_RATE,
weight_decay=WEIGHT_DECAY)
    scheduler = optim.lr_scheduler.OneCycleLR(
        optimizer,
        max_lr=LEARNING_RATE,
        epochs=EPOCHS,
        steps_per_epoch=len(train_loader),
        pct_start=0.15,
    )
    criterion = nn.CrossEntropyLoss(label_smoothing=0.05)
    scaler = torch.amp.GradScaler("cuda", enabled=device.type ==
"cuda")

    best_acc = 0.0
    best_true = []
    best_pred = []
    started = time.perf_counter()

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    for epoch in range(1, EPOCHS + 1):
        train_loss, train_acc = train_one_epoch(model, train_loader,
        criterion, optimizer, scheduler, scaler, device)
        test_loss, test_acc, y_true, y_pred = evaluate(model,
        test_loader, criterion, device)
        if test_acc >= best_acc:
            best_acc = test_acc
            best_true = y_true
            best_pred = y_pred
        print(
            f"epoch {epoch:02d}/{EPOCHS} "
            f"train_loss={train_loss:.4f} train_acc={train_acc:.4f} "
            f"test_loss={test_loss:.4f} test_acc={test_acc:.4f}"
        )

    return {
        "model": "Fast14M_Residual_LeNetStyle_CNN",
        "parameters": param_count,
        "accuracy": best_acc,
        "train_seconds": round(time.perf_counter() - started, 2),
        "status": "ok",
        "y_true": best_true,
        "y_pred": best_pred,
    }

def main() -> None:
    root = upload_and_extract_zip()
    seed_everything(SEED)
    OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

    device = torch.device("cuda" if torch.cuda.is_available() else
    "cpu")
    print("Device:", device)
    if device.type != "cuda":
        print("Warning: GPU not detected. This hybrid is heavy and
        will be slow on CPU.")

    train_dir = find_split_dir(root, "train")
    test_dir = find_split_dir(root, "test")

    print("\nDataset folders")
    print("Train:", train_dir)
    print("Test: ", test_dir)
    print("Train image count:", count_images(train_dir))
    print("Test image count: ", count_images(test_dir))

    train_tfms, test_tfms = get_transforms()
    train_dataset = LimitedImageFolder(train_dir, train_tfms,

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MAX_TRAIN_IMAGES)
    test_dataset = LimitedImageFolder(test_dir, test_tfms,
MAX_TEST_IMAGES)

    if train_dataset.classes != test_dataset.classes:
        print("Warning: train/test class folders differ.")
        print("Train classes:", train_dataset.classes)
        print("Test classes: ", test_dataset.classes)

    num_classes = len(train_dataset.classes)
    print("\nClasses:", num_classes)
    print("Train samples used:", len(train_dataset))
    print("Test samples used: ", len(test_dataset))
    print("\nFast path")
    print("64x64 image -> LeNet-style stem -> residual stages -> MLP
mixer -> classifier")

    train_loader = DataLoader(
        train_dataset,
        batch_size=BATCH_SIZE,
        shuffle=True,
        num_workers=NUM_WORKERS,
        pin_memory=torch.cuda.is_available(),
        persistent_workers=NUM_WORKERS > 0,
    )
    test_loader = DataLoader(
        test_dataset,
        batch_size=BATCH_SIZE,
        shuffle=False,
        num_workers=NUM_WORKERS,
        pin_memory=torch.cuda.is_available(),
        persistent_workers=NUM_WORKERS > 0,
    )

    payload = fit_model(num_classes, train_loader, test_loader,
device)
    leaderboard = pd.DataFrame([k: v for k, v in payload.items() if k
not in ["y_true", "y_pred"]])
    leaderboard.insert(0, "rank", [1])

    print("\nLeaderboard")
    display(leaderboard)

    leaderboard_path = OUTPUT_DIR /
"fast14m_character_leaderboard.csv"
    leaderboard.to_csv(leaderboard_path, index=False)
    print("Saved leaderboard:", leaderboard_path)

    class_names = train_dataset.classes
    y_true_names = [class_names[i] for i in payload["y_true"]]

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y_pred_names = [class_names[i] for i in payload["y_pred"]]

print(f"\nBest model: {payload['model']}")
print(f"Best accuracy: {payload['accuracy']:.4f}")
print("\nClassification report")
print(classification_report(y_true_names, y_pred_names,
zero_division=0))

cm = pd.DataFrame(
    confusion_matrix(y_true_names, y_pred_names,
labels=class_names),
    index=class_names,
    columns=class_names,
)
print("\nConfusion matrix")
display(cm)

cm_path = OUTPUT_DIR / "best_model_confusion_matrix.csv"
cm.to_csv(cm_path)
print("Saved confusion matrix:", cm_path)
files.download(str(leaderboard_path))

if __name__ == "__main__":
    main()

```

Upload your train/test zip file now.

If the upload button does not appear, upload the zip from Colab's left Files panel and rerun.

<IPython.core.display.HTML object>

Saving Dataset.zip to Dataset.zip

Extracted to: /content/bengali_hybrid_dataset

Device: cuda

Dataset folders

Train: /content/bengali_hybrid_dataset/Dataset/train

Test: /content/bengali_hybrid_dataset/Dataset/test

Train image count: 12000

Test image count: 3000

Classes: 50

Train samples used: 12000

Test samples used: 3000

Fast path

64x64 image -> LeNet-style stem -> residual stages -> MLP mixer -> classifier

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/
dataloader.py:424: UserWarning: This DataLoader will create 4 worker
processes in total. Our suggested max number of worker in current
system is 2, which is smaller than what this DataLoader is going to
create. Please be aware that excessive worker creation might get
DataLoader running slow or even freeze, lower the worker number to
avoid potential slowness/freeze if necessary.
  self.check_worker_number_rationality()
```

torch.compile enabled

Trainable parameters: 14,474,866 (1.45 crore)

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/
dataloader.py:432: UserWarning: This DataLoader will create 4 worker
processes in total. Our suggested max number of worker in current
system is 2, which is smaller than what this DataLoader is going to
create. Please be aware that excessive worker creation might get
DataLoader running slow or even freeze, lower the worker number to
avoid potential slowness/freeze if necessary.
```

```
  self.check_worker_number_rationality()
```

```
W0530 21:23:24.014000 5443 torch/_inductor/utils.py:1731] [0/0] Not
enough SMs to use max_autotune_gemm mode
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py
:432: UserWarning: This DataLoader will create 4 worker processes in
total. Our suggested max number of worker in current system is 2,
which is smaller than what this DataLoader is going to create. Please
be aware that excessive worker creation might get DataLoader running
slow or even freeze, lower the worker number to avoid potential
slowness/freeze if necessary.
```

```
  self.check_worker_number_rationality()
```

```
epoch 01/20 train_loss=3.6631 train_acc=0.0634 test_loss=9.5764
test_acc=0.0333
```

```
epoch 02/20 train_loss=2.6803 train_acc=0.2555 test_loss=3.6602
test_acc=0.1553
```

```
epoch 03/20 train_loss=1.5614 train_acc=0.5970 test_loss=1.3520
test_acc=0.6830
```

```
epoch 04/20 train_loss=0.9604 train_acc=0.8219 test_loss=1.1232
test_acc=0.7540
```

```
epoch 05/20 train_loss=0.7261 train_acc=0.9003 test_loss=0.6978
test_acc=0.9180
```

```
epoch 06/20 train_loss=0.6344 train_acc=0.9314 test_loss=0.5868
test_acc=0.9503
```

```
epoch 07/20 train_loss=0.5645 train_acc=0.9543 test_loss=0.6384
test_acc=0.9193
```

```
epoch 08/20 train_loss=0.5256 train_acc=0.9672 test_loss=0.5614
test_acc=0.9527
```

```
epoch 09/20 train_loss=0.4947 train_acc=0.9744 test_loss=0.5242
test_acc=0.9673
```

```
epoch 10/20 train_loss=0.4721 train_acc=0.9825 test_loss=0.4757
```

```

test_acc=0.9770
epoch 11/20 train_loss=0.4551 train_acc=0.9868 test_loss=0.5011
test_acc=0.9703
epoch 12/20 train_loss=0.4471 train_acc=0.9888 test_loss=0.4832
test_acc=0.9767
epoch 13/20 train_loss=0.4370 train_acc=0.9916 test_loss=0.4557
test_acc=0.9843
epoch 14/20 train_loss=0.4262 train_acc=0.9935 test_loss=0.4523
test_acc=0.9837
epoch 15/20 train_loss=0.4209 train_acc=0.9950 test_loss=0.4503
test_acc=0.9820
epoch 16/20 train_loss=0.4170 train_acc=0.9961 test_loss=0.4425
test_acc=0.9857
epoch 17/20 train_loss=0.4139 train_acc=0.9967 test_loss=0.4424
test_acc=0.9853
epoch 18/20 train_loss=0.4123 train_acc=0.9968 test_loss=0.4420
test_acc=0.9857
epoch 19/20 train_loss=0.4121 train_acc=0.9970 test_loss=0.4403
test_acc=0.9853
epoch 20/20 train_loss=0.4105 train_acc=0.9976 test_loss=0.4402
test_acc=0.9857

```

Leaderboard

```

{"summary":{"\n  \"name\": \"    main()\", \n  \"rows\": 1, \n  \"fields\": [\n    {\n      \"column\": \"rank\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": null, \n        \"min\": 1, \n        \"max\": 1, \n        \"num_unique_values\": 1, \n        \"samples\": [\n          1\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      {\n        \"column\": \"model\", \n        \"properties\": {\n          \"dtype\": \"string\", \n          \"num_unique_values\": 1, \n          \"samples\": [\n            \"Fast14M_Residual_LeNetStyle_CNN\"\n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\"\n        }, \n        {\n          \"column\": \"parameters\", \n          \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": null, \n            \"min\": 14474866, \n            \"max\": 14474866, \n            \"num_unique_values\": 1, \n            \"samples\": [\n              14474866\n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\"\n          }, \n          {\n            \"column\": \"accuracy\", \n            \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": null, \n              \"min\": 0.9856666666666667, \n              \"max\": 0.9856666666666667, \n              \"num_unique_values\": 1, \n              \"samples\": [\n                0.9856666666666667\n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\"\n            }, \n            {\n              \"column\": \"train_seconds\", \n              \"properties\": {\n                \"dtype\": \"number\", \n                \"std\": null, \n                \"min\": 505.56, \n                \"max\": 505.56, \n                \"num_unique_values\": 1, \n                \"samples\": [\n                  505.56\n                ], \n

```

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{"semantic_type": "\n", "description": "\n", "column": "status", "properties": {"dtype": "string", "num_unique_values": 1, "samples": [{"ok"}]}, "semantic_type": "\n", "description": "\n", "column": "status", "properties": {"dtype": "string", "num_unique_values": 1, "samples": [{"ok"}]}, "type": "dataframe"}

```

Saved leaderboard:

/content/bengali_hybrid_results/fast14m_character_leaderboard.csv

Best model: Fast14M_Residual_LeNetStyle_CNN

Best accuracy: 0.9857

Classification report

	precision	recall	f1-score	support
172	0.98	0.98	0.98	60
173	0.98	1.00	0.99	60
174	1.00	0.95	0.97	60
175	0.98	1.00	0.99	60
176	0.95	0.98	0.97	60
177	0.98	1.00	0.99	60
178	1.00	0.98	0.99	60
179	1.00	0.98	0.99	60
180	0.98	1.00	0.99	60
181	1.00	1.00	1.00	60
182	1.00	1.00	1.00	60
183	0.98	0.98	0.98	60
184	0.98	0.98	0.98	60
185	0.98	1.00	0.99	60
186	0.97	0.97	0.97	60
187	1.00	0.98	0.99	60
188	0.98	1.00	0.99	60
189	0.98	1.00	0.99	60
190	0.98	1.00	0.99	60
191	0.98	1.00	0.99	60
192	0.98	0.98	0.98	60
193	0.98	1.00	0.99	60
194	1.00	0.98	0.99	60
195	1.00	0.98	0.99	60
196	1.00	0.98	0.99	60
197	0.92	0.97	0.94	60
198	0.95	0.97	0.96	60
199	0.95	0.98	0.97	60
200	1.00	1.00	1.00	60
201	0.98	0.97	0.97	60
202	0.96	0.90	0.93	60
203	1.00	0.98	0.99	60
204	0.98	0.98	0.98	60
205	1.00	0.98	0.99	60

206	1.00	1.00	1.00	60
207	0.97	1.00	0.98	60
208	0.97	0.93	0.95	60
209	1.00	0.98	0.99	60
210	1.00	1.00	1.00	60
211	0.98	1.00	0.99	60
212	0.98	1.00	0.99	60
213	1.00	0.95	0.97	60
214	0.98	1.00	0.99	60
215	1.00	1.00	1.00	60
216	1.00	0.98	0.99	60
217	1.00	1.00	1.00	60
218	1.00	1.00	1.00	60
219	0.98	1.00	0.99	60
220	1.00	1.00	1.00	60
221	0.98	0.97	0.97	60
accuracy			0.99	3000
macro avg	0.99	0.99	0.99	3000
weighted avg	0.99	0.99	0.99	3000

Confusion matrix

```
{"type": "dataframe"}
```

Saved confusion matrix:

/content/bengali_hybrid_results/best_model_confusion_matrix.csv

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

```
import os
import time
import zipfile
import json
from pathlib import Path
from typing import Dict, List, Tuple, Any
import numpy as np
import pandas as pd
from PIL import Image
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix, precision_recall_fscore_support

try:
    from google.colab import files
```

```

    IN_COLAB = True
except ImportError:
    IN_COLAB = False

# ENGINE CONFIGURATIONS
ZIP_LIMIT_MB = 2000
DATA_DIR = Path("/content/bengali_character_data") if IN_COLAB else Path("./bengali_character_data")
IMAGE_EXTENSIONS = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}

IMAGE_SIZE = 32
BATCH_SIZE = 256 # Maximizes GPU parallelization performance
EPOCHS = 20
LEARNING_RATE = 0.003
RANDOM_STATE = 42

torch.manual_seed(RANDOM_STATE)
np.random.seed(RANDOM_STATE)

if torch.cuda.is_available():
    device = torch.device("cuda")
    torch.backends.cudnn.benchmark = True
else:
    device = torch.device("cpu")
print(f" Execution Core: {device}")

def upload_and_extract_zip() -> Path:
    """
    Optimized data engine: Bypasses extraction if dataset is present.
    Safely captures downstream target sub-folders.
    """
    if DATA_DIR.exists() and any(DATA_DIR.iterdir()):
        print(f"> [SKIPPED] Active dataset directory detected at '{DATA_DIR}'. Bypassing upload/extraction completely.")
        return DATA_DIR

    zip_path = None
    if IN_COLAB:
        content_zips = list(Path("/content").glob("*.zip"))
        if content_zips:
            zip_path = sorted(content_zips, key=lambda x:
x.stat().st_mtime)[-1]
            print(f" Detected archive in Colab workspace storage: {zip_path.name}")
        else:
            print(" [STEP 1] Please select and upload your dataset zip file...")
            uploaded = files.upload()
            if not uploaded:
                raise RuntimeError("X Upload aborted by user.")

```

```

        zip_name = next(iter(uploaded.keys()))
        zip_path = Path("/content") / zip_name
    else:
        print(" Local Execution Mode. Searching for dataset.zip...")
        zip_path = Path("dataset.zip")
        if not zip_path.exists():
            zips = list(Path(".").glob("*.zip"))
            if zips: zip_path = zips[0]
            else: raise FileNotFoundError("X Missing local zip
archive.")

        print(f" Unzipping archive: {zip_path} ->
{DATA_DIR.resolve()}...")
        DATA_DIR.mkdir(parents=True, exist_ok=True)

        with zipfile.ZipFile(zip_path) as archive:
            for member in archive.infolist():
                target = (DATA_DIR / member.filename).resolve()
                if not str(target).startswith(str(DATA_DIR.resolve())):
                    raise ValueError("X Security violation: Dangerous
relative path pattern inside zip file.")
                archive.extractall(DATA_DIR)

        print(" Extraction successfully completed.")
        return DATA_DIR

class BengaliDataset(Dataset):
    """
    Fault-tolerant character dataset loader.
    Validates structural files during initialization to eliminate
    DataLoader runtime crashes.
    """
    def __init__(self, folder_path: Path, class_names: List[str],
class_to_idx: Dict[str, int]):
        self.samples = []
        self.class_to_idx = class_to_idx

        print(f" Inspecting and sanitizing data files in
{folder_path.name}...")
        corrupt_counter = 0

        for class_name in class_names:
            class_folder = folder_path / class_name
            if not class_folder.exists():
                continue
            idx = class_to_idx[class_name]

            for file_path in class_folder.iterdir():
                if file_path.is_file() and file_path.suffix.lower() in
IMAGE_EXTENSIONS:

```

```

        # PRO FIX: Fast structural verification check to
        filter corrupted assets early
        if file_path.stat().st_size == 0:
            corrupt_counter += 1
            continue
        try:
            with Image.open(file_path) as img:
                img.verify() # Validates internal byte
structure without fully reading matrix maps
            self.samples.append((file_path, idx))
        except Exception:
            corrupt_counter += 1
            continue

    if corrupt_counter > 0:
        print(f"⚠ Cleaned up {corrupt_counter}
unidentifiable/corrupt file assets from execution buffers.")

    def __len__(self):
        return len(self.samples)

    def __getitem__(self, item):
        file_path, label = self.samples[item]
        try:
            with Image.open(file_path) as img:
                img = img.convert("L")
                img = img.resize((IMAGE_SIZE, IMAGE_SIZE),
Image.Resampling.BILINEAR)
                arr = np.array(img, dtype=np.float32) / 255.0
                tensor = torch.tensor(arr,
dtype=torch.float32).unsqueeze(0)
                tensor = (tensor - 0.5) / 0.5 # Scale normalization
to range [-1, 1]
            return tensor, label
        except Exception:
            # Fallback block: Return a zero tensor if an operational
            file system lockup or rare read crash occurs
            return torch.zeros((1, IMAGE_SIZE, IMAGE_SIZE),
dtype=torch.float32), label

#
=====
===
# SYSTEM SCALED TWO-STAGE CASCADED PIPELINE (MAXIMIZED FOR ~195K
PARAMS)
#
=====
===

class Stage1CoarseClassifier(nn.Module):

```

```

    """Stage 1: Scaled to extract distinct structural global feature
    maps."""
    def __init__(self, num_classes: int):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(1, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2), # 32 x 16 x 16
            nn.Conv2d(32, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2), # 64 x 8 x 8
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128), nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((1, 1)) # 128 features
        )
        self.classifier = nn.Linear(128, num_classes)

    def forward(self, x):
        x = self.features(x)
        x = x.view(x.size(0), -1)
        return self.classifier(x)

class Stage2ExtremalRefiner(nn.Module):
    """Stage 2: Consumes raw spatial features alongside Stage 1 logit
    predictions."""
    def __init__(self, num_classes: int):
        super().__init__()
        self.img_features = nn.Sequential(
            nn.Conv2d(1, 16, kernel_size=3, padding=1),
            nn.BatchNorm2d(16), nn.ReLU(inplace=True),
            nn.MaxPool2d(4, 4), # 16 x 8 x 8
            nn.Conv2d(16, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32), nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((1, 1)) # 32 features
        )
        # Optimized bottleneck layers for precise boundary adjustments
        self.refiner = nn.Sequential(
            nn.Linear(32 + num_classes, 128),
            nn.BatchNorm1d(128),
            nn.ReLU(inplace=True),
            nn.Dropout(0.2),
            nn.Linear(128, num_classes)
        )

    def forward(self, x, stage1_logits):
        img_feats = self.img_features(x)
        img_feats = img_feats.view(img_feats.size(0), -1)
        combined = torch.cat([img_feats, stage1_logits], dim=1)
        return self.refiner(combined)

```

```

class Cascaded99Pipeline(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.stage1 = Stage1CoarseClassifier(num_classes)
        self.stage2 = Stage2ExtremalRefiner(num_classes)

    def forward(self, x):
        s1_out = self.stage1(x)
        s2_out = self.stage2(x, s1_out)
        return s2_out

def generate_html_report(history: Dict[str, List[float]], metrics:
Dict[str, Any], total_params: int) -> str:
    history_json = json.dumps(history)
    report_cleaned = "<br>".join(metrics["class_report"].split("\n"))
    accuracy = metrics["accuracy"] * 100
    return f"""<!DOCTYPE html>
<html lang="en" class="dark">
<head>
    <meta charset="UTF-8"><title>Extreme Pipeline
Analytics</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>body {{ background: #030712; color: #f3f4f6; font-
family: sans-serif; }}</style>
</head>
<body class="p-12">
    <header class="max-w-7xl mx-auto mb-10 flex justify-between
border-b border-gray-800 pb-6">
        <div><h1 class="text-3xl font-extrabold text-
white">Cascaded Optimization System Dashboard</h1></div>
        <div class="bg-gray-900 px-6 py-2 rounded-xl text-
center"><span class="text-3xl font-extrabold text-emerald-
400">{accuracy:.2f}%</span></div>
    </header>
    <main class="max-w-7xl mx-auto space-y-8">
        <div class="bg-gray-900 p-6 rounded-xl"><p class="text-
amber-400">Total Validated Parameters: {total_params:,} /
200,000</p></div>
        <div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="lossChart" height="250"></canvas></div>
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="accuracyChart" height="250"></canvas></div>
        </div>
        <div class="bg-gray-900 p-6 rounded-xl"><pre class="text-
xs text-cyan-400 font-mono">{report_cleaned}</pre></div>
    </main>
</body>
</html>
    """

```

```

        const data = {history_json};
        new Chart(document.getElementById('lossChart'), {{ type:
'line', data: {{ labels: Array.from({{length: data.loss.length}},
(_,i)=>i+1), datasets: [{{label:'Loss', data:data.loss,
borderColor:'#f87171', tension:0.1}}] }} }});
        new Chart(document.getElementById('accuracyChart'),
{{ type: 'line', data: {{ labels: Array.from({{length:
data.acc.length}}, (_,i)=>i+1), datasets: [{{label:'Accuracy',
data:data.acc.map(a=>a*100), borderColor:'#34d399', tension:0.1}}] }}
}});
    </script>
</body></html>""

```

```

#
=====
===
# MASTER PIPELINE ENGINE RUNNER
#
=====
===
def main():
    print("=====")
    print(" Executing Two-Stage Cascaded Extremal Pipeline Engine ")
    print("=====")

    try:
        data_path = upload_and_extract_zip()
    except Exception as e:
        print(f"⚠ Runtime Exception Intercepted: {e}")
        return

    # Dynamic lookups to accommodate deep extraction folder nesting
    # profiles
    train_dir, test_dir = data_path / "train", data_path / "test"
    if not train_dir.exists():
        # Look for subdirectories containing 'train' if nested under
        # another top-level directory
        for subdir in data_path.rglob("train"):
            if subdir.is_dir():
                train_dir = subdir
                test_dir = subdir.parent / "test"
                break
    else:
        # Fallback pathing structure parsing
        for subdir in data_path.iterdir():
            if subdir.is_dir() and (subdir / "train").exists():
                train_dir, test_dir = subdir / "train", subdir /
"test"
                break
        else:

```

```

        raise FileNotFoundError("X Structural anomaly: Could
not locate 'train' and 'test' subdirectories.")

    class_names = sorted([d.name for d in train_dir.iterdir() if
d.is_dir()])
    class_to_idx = {name: idx for idx, name in enumerate(class_names)}
    num_classes = len(class_names)

    print(f" Target Classes Discovered: {num_classes}")

    # Run data verification matrices
    train_dataset = BengaliDataset(train_dir, class_names,
class_to_idx)
    test_dataset = BengaliDataset(test_dir, class_names, class_to_idx)

    # PRO FIX: Lowered worker count to 2 to eliminate multi-processing
system overhead and warnings
    train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE,
shuffle=True, num_workers=2, pin_memory=True)
    test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)

    pipeline = Cascaded99Pipeline(num_classes=num_classes).to(device)

    # Native compilation layer configuration
    if hasattr(torch, 'compile') and device.type == 'cuda':
        try:
            print("\n Compiling graph architectures via
torch.compile()...")
            pipeline = torch.compile(pipeline)
        except Exception:
            print(" Standard sequential computation mode selected.")

    total_params = sum(p.numel() for p in pipeline.parameters() if
p.requires_grad)
    print(f" Combined Pipeline Footprint: {total_params:,} Trainable
Parameters Initialization.")
    if total_params > 200000:
        print(f"⚠ Warning: Total parameter size ({total_params})
breaks strict 200K limit.")

    criterion = nn.CrossEntropyLoss()

    opt_s1 = optim.AdamW(pipeline.stage1.parameters(),
lr=LEARNING_RATE, weight_decay=1e-2, fused=True if device.type=='cuda'
else False)
    opt_s2 = optim.AdamW(pipeline.stage2.parameters(),
lr=LEARNING_RATE, weight_decay=1e-2, fused=True if device.type=='cuda'
else False)

```

```

    sched_s1 = optim.lr_scheduler.OneCycleLR(opt_s1,
max_lr=LEARNING_RATE, steps_per_epoch=len(train_loader),
epochs=EPOCHS)
    sched_s2 = optim.lr_scheduler.OneCycleLR(opt_s2,
max_lr=LEARNING_RATE, steps_per_epoch=len(train_loader),
epochs=EPOCHS)

    scaler = torch.amp.GradScaler('cuda' if device.type == 'cuda' else
'cpu')

    history = {"loss": [], "acc": []}

    print(f"\n Commencing Backpropagation Pipeline Over {EPOCHS}
Faster Cycles...")
    for epoch in range(EPOCHS):
        start = time.time()
        pipeline.train()
        running_loss, correct, total = 0.0, 0, 0

        for images, labels in train_loader:
            images, labels = images.to(device, non_blocking=True),
labels.to(device, non_blocking=True)

            opt_s1.zero_grad(set_to_none=True)
            opt_s2.zero_grad(set_to_none=True)

            with torch.amp.autocast('cuda' if device.type == 'cuda'
else 'cpu'):
                s1_logits = pipeline.stage1(images)
                s2_logits = pipeline.stage2(images,
s1_logits.detach())

                loss_s1 = criterion(s1_logits, labels)
                loss_s2 = criterion(s2_logits, labels)
                combined_loss = loss_s1 + loss_s2

            scaler.scale(combined_loss).backward()

            scaler.step(opt_s1)
            scaler.step(opt_s2)
            scaler.update()

            sched_s1.step()
            sched_s2.step()

            running_loss += loss_s2.item() * images.size(0)
            _, predicted = torch.max(s2_logits.data, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()

```

```

        epoch_loss = running_loss / max(1, len(train_dataset))
        epoch_acc = correct / max(1, total)
        history["loss"].append(epoch_loss)
        history["acc"].append(epoch_acc)

        print(f"Cycle [{epoch+1}/{EPOCHS}] ({time.time()-start:.1f}s)
-> Pipeline Loss: {epoch_loss:.4f} | Target Acc: {epoch_acc*100:.2f}
%")

        if epoch_acc >= 0.993:
            print(" Target convergence criteria achieved (>= 99.3%).
Freezing learning loop pipeline layers.")
            break

    # Evaluate final metrics
    pipeline.eval()
    all_preds, all_labels = [], []
    with torch.no_grad():
        for images, labels in test_loader:
            images = images.to(device)
            preds = pipeline(images)
            _, predicted = torch.max(preds, 1)
            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.numpy())

    all_preds, all_labels = np.array(all_preds), np.array(all_labels)
    acc = accuracy_score(all_labels, all_preds)
    p, r, _, _ = precision_recall_fscore_support(all_labels,
all_preds, average="weighted", zero_division=0)
    class_rep = classification_report(all_labels, all_preds,
target_names=class_names, zero_division=0)

    metrics = {"accuracy": float(acc), "precision_weighted": float(p),
"recall_weighted": float(r), "class_report": class_rep}

    html_report = generate_html_report(history, metrics, total_params)
    with open("cascaded_extreme_report.html", "w", encoding="utf-8")
as f: f.write(html_report)
    if IN_COLAB: files.download("cascaded_extreme_report.html")
    print("\n Pipeline execution completed successfully.")

if __name__ == "__main__":
    main()

```

Execution Core: cuda

```

=====
Executing Two-Stage Cascaded Extremal Pipeline Engine
=====

```

```

< [SKIPPED] Active dataset directory detected at
'/content/bengali_character_data'. Bypassing upload/extraction

```

completely.

Target Classes Discovered: 50

Inspecting and sanitizing data files in train...

△ Cleaned up 1 unidentifiable/corrupt file assets from execution buffers.

Inspecting and sanitizing data files in test...

□ Compiling graph architectures via torch.compile()...

Combined Pipeline Footprint: 121,796 Trainable Parameters

Initialization.

Commencing Backpropagation Pipeline Over 20 Faster Cycles...

Cycle [1/20]	(22.6s)	-> Pipeline Loss: 3.8814	Target Acc: 3.88%
Cycle [2/20]	(3.0s)	-> Pipeline Loss: 3.5361	Target Acc: 10.24%
Cycle [3/20]	(3.4s)	-> Pipeline Loss: 2.8550	Target Acc: 24.33%
Cycle [4/20]	(3.9s)	-> Pipeline Loss: 2.0999	Target Acc: 41.11%
Cycle [5/20]	(3.1s)	-> Pipeline Loss: 1.5650	Target Acc: 52.70%
Cycle [6/20]	(3.0s)	-> Pipeline Loss: 1.2099	Target Acc: 62.92%
Cycle [7/20]	(3.5s)	-> Pipeline Loss: 0.9485	Target Acc: 70.64%
Cycle [8/20]	(3.7s)	-> Pipeline Loss: 0.7378	Target Acc: 76.60%
Cycle [9/20]	(2.9s)	-> Pipeline Loss: 0.6024	Target Acc: 81.31%
Cycle [10/20]	(3.0s)	-> Pipeline Loss: 0.5184	Target Acc: 83.80%
Cycle [11/20]	(3.2s)	-> Pipeline Loss: 0.4541	Target Acc: 85.82%
Cycle [12/20]	(4.1s)	-> Pipeline Loss: 0.3711	Target Acc: 88.59%
Cycle [13/20]	(3.0s)	-> Pipeline Loss: 0.3250	Target Acc: 89.69%
Cycle [14/20]	(3.1s)	-> Pipeline Loss: 0.2915	Target Acc: 91.27%
Cycle [15/20]	(3.2s)	-> Pipeline Loss: 0.2587	Target Acc: 92.49%
Cycle [16/20]	(4.1s)	-> Pipeline Loss: 0.2342	Target Acc: 92.89%
Cycle [17/20]	(3.0s)	-> Pipeline Loss: 0.2091	Target Acc: 93.60%
Cycle [18/20]	(3.0s)	-> Pipeline Loss: 0.1967	Target Acc: 94.55%
Cycle [19/20]	(3.3s)	-> Pipeline Loss: 0.1910	Target Acc: 94.87%
Cycle [20/20]	(4.1s)	-> Pipeline Loss: 0.1881	Target Acc: 94.62%

W0531 06:21:12.124000 1116 torch/_inductor/utils.py:1731] [0/0] Not enough SMs to use max_autotune_gemm mode

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

Pipeline execution completed successfully.

```
import os
import time
import zipfile
import json
from pathlib import Path
from typing import Dict, List, Tuple, Any
import numpy as np
import pandas as pd
from PIL import Image, ImageOps
```

```

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import accuracy_score, classification_report,
precision_recall_fscore_support

try:
    from google.colab import files
    IN_COLAB = True
except ImportError:
    IN_COLAB = False

# ENGINE CONFIGURATIONS
DATA_DIR = Path("/content/bengali_character_data") if IN_COLAB else
Path("./bengali_character_data")
IMAGE_EXTENSIONS = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}

IMAGE_SIZE = 32
BATCH_SIZE = 128 # Slightly tighter batch size for more gradient
updates per epoch
EPOCHS = 35 # Increased capacity space for complete convergence
LEARNING_RATE = 0.002
RANDOM_STATE = 42

torch.manual_seed(RANDOM_STATE)
np.random.seed(RANDOM_STATE)

if torch.cuda.is_available():
    device = torch.device("cuda")
    torch.backends.cudnn.benchmark = True
else:
    device = torch.device("cpu")
print(f" Execution Core: {device}")

def upload_and_extract_zip() -> Path:
    """
    Looks for the newest uploaded archive zip file in workspace
    storage.
    If a new file is detected, it wipes prior cache to avoid data
    pollution.
    """
    zip_path = None
    if IN_COLAB:
        content_zips = list(Path("/content").glob("*.zip"))
        if content_zips:
            # Sort to grab the absolute newest upload in the workspace
            zip_path = sorted(content_zips, key=lambda x:
x.stat().st_mtime)[-1]
            print(f" Detected target archive for extraction:

```

```

{zip_path.name}")
    else:
        print(" [STEP 1] Please select and upload your dataset zip
file...")
        uploaded = files.upload()
        if not uploaded:
            raise RuntimeError("X Upload aborted by user.")
        zip_name = next(iter(uploaded.keys()))
        zip_path = Path("/content") / zip_name
    else:
        zip_path = Path("dataset.zip")
        if not zip_path.exists():
            zips = list(Path(".").glob("*.zip"))
            if zips: zip_path = zips[0]
            else: raise FileNotFoundError("X Missing local zip
archive.")

    # Track if we have extracted data already matching this zip
lifecycle
    marker_file = DATA_DIR / f".extracted_{zip_path.stem}"
    if DATA_DIR.exists() and marker_file.exists():
        print(f"> [SKIPPED] Pre-extracted cache matches target
archive. Proceeding directly to dataloaders.")
        return DATA_DIR

    # Clear directory if a fresh zip was uploaded to prevent class
cross-pollution
    if DATA_DIR.exists():
        import shutil
        print(" Wiping outdated data cache folder...")
        shutil.rmtree(DATA_DIR)

    print(f" Unzipping fresh archive: {zip_path} ->
{DATA_DIR.resolve()}...")
    DATA_DIR.mkdir(parents=True, exist_ok=True)

    with zipfile.ZipFile(zip_path) as archive:
        for member in archive.infolist():
            target = (DATA_DIR / member.filename).resolve()
            if not str(target).startswith(str(DATA_DIR.resolve())):
                raise ValueError("X Security violation: Dangerous
relative path pattern inside zip file.")
            archive.extractall(DATA_DIR)

    # Drop marker file to avoid reprocessing during this runtime
session
    marker_file.touch()
    print(" Extraction successfully completed.")
    return DATA_DIR

```

```

class BengaliDataset(Dataset):
    """
    Fault-tolerant character dataset loader featuring localized
    deterministic data
    augmentations to prevent overfitting on complex handwriting
    geometries.
    """
    def __init__(self, folder_path: Path, class_names: List[str],
class_to_idx: Dict[str, int], is_train: bool = False):
        self.samples = []
        self.class_to_idx = class_to_idx
        self.is_train = is_train

        print(f" Validating and caching assets from {folder_path.name}
partitions...")
        corrupt_counter = 0

        for class_name in class_names:
            class_folder = folder_path / class_name
            if not class_folder.exists(): continue
            idx = class_to_idx[class_name]

            for file_path in class_folder.iterdir():
                if file_path.is_file() and file_path.suffix.lower() in
IMAGE_EXTENSIONS:
                    if file_path.stat().st_size == 0:
                        corrupt_counter += 1
                        continue
                    try:
                        with Image.open(file_path) as img:
                            img.verify()
                            self.samples.append((file_path, idx))
                    except Exception:
                        corrupt_counter += 1
                        continue

            if corrupt_counter > 0:
                print(f"△ Safely filtered out {corrupt_counter} corrupted
image assets from execution streams.")

        def __len__(self):
            return len(self.samples)

        def __getitem__(self, item):
            file_path, label = self.samples[item]
            try:
                with Image.open(file_path) as img:
                    img = img.convert("L")

                # Apply high-speed training runtime transforms

```

```

        if self.is_train:
            if np.random.rand() > 0.4:
                # Controlled random rotation mapping for
                character skew resilience
                angle = np.random.uniform(-10, 10)
                img = img.rotate(angle,
resample=Image.Resampling.BILINEAR)

                img = img.resize((IMAGE_SIZE, IMAGE_SIZE),
Image.Resampling.BILINEAR)
                arr = np.array(img, dtype=np.float32) / 255.0
                tensor = torch.tensor(arr,
dtype=torch.float32).unsqueeze(0)
                tensor = (tensor - 0.5) / 0.5
                return tensor, label
            except Exception:
                return torch.zeros((1, IMAGE_SIZE, IMAGE_SIZE),
dtype=torch.float32), label

#
=====
==
# MAXIMUM CAPACITY TWO-STAGE CASCADED PIPELINE (~191K PARAMS)
#
=====
==

class Stage1CoarseClassifier(nn.Module):
    """Stage 1: Optimized feature extractor avoiding aggressive
    spatial compression."""
    def __init__(self, num_classes: int):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(1, 32, kernel_size=3, padding=1),
nn.BatchNorm2d(32), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2), # 32 x 16 x 16

            nn.Conv2d(32, 64, kernel_size=3, padding=1),
nn.BatchNorm2d(64), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2), # 64 x 8 x 8

            nn.Conv2d(64, 96, kernel_size=3, padding=1),
nn.BatchNorm2d(96), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2) # 96 x 4 x 4
        )
        self.flat_dim = 96 * 4 * 4
        self.classifier = nn.Linear(self.flat_dim, num_classes)

    def forward(self, x):
        x = self.features(x)

```

```

        x = x.view(x.size(0), -1)
        return self.classifier(x)

class Stage2ExtremalRefiner(nn.Module):
    """Stage 2: Cross-references structural feature maps with Stage
    1's distribution logit vectors."""
    def __init__(self, num_classes: int):
        super().__init__()
        self.img_features = nn.Sequential(
            nn.Conv2d(1, 16, kernel_size=3, padding=1),
            nn.BatchNorm2d(16), nn.ReLU(inplace=True),
            nn.MaxPool2d(4, 4), # 16 x 8 x 8
            nn.Conv2d(16, 24, kernel_size=3, padding=1),
            nn.BatchNorm2d(24), nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((2, 2)) # 24 x 2 x 2 = 96 features
        )
        self.refiner = nn.Sequential(
            nn.Linear(96 + num_classes, 128),
            nn.BatchNorm1d(128),
            nn.ReLU(inplace=True),
            nn.Dropout(0.15),
            nn.Linear(128, num_classes)
        )

    def forward(self, x, stage1_logits):
        img_feats = self.img_features(x)
        img_feats = img_feats.view(img_feats.size(0), -1)
        combined = torch.cat([img_feats, stage1_logits], dim=1)
        return self.refiner(combined)

class Cascaded99Pipeline(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.stage1 = Stage1CoarseClassifier(num_classes)
        self.stage2 = Stage2ExtremalRefiner(num_classes)

    def forward(self, x):
        s1_out = self.stage1(x)
        s2_out = self.stage2(x, s1_out)
        return s2_out

def generate_html_report(history: Dict[str, List[float]], metrics:
Dict[str, Any], total_params: int) -> str:
    history_json = json.dumps(history)
    report_cleaned = "<br>".join(metrics["class_report"].split("\n"))
    accuracy = metrics["accuracy"] * 100
    return f"""<!DOCTYPE html>
<html lang="en" class="dark">
<head>
    <meta charset="UTF-8"><title>Extreme Pipeline

```

```

Analytics</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>body {{ background: #030712; color: #f3f4f6; font-
family: sans-serif; }}</style>
</head>
<body class="p-12">
    <header class="max-w-7xl mx-auto mb-10 flex justify-between
border-b border-gray-800 pb-6">
        <div><h1 class="text-3xl font-extrabold text-
white">Cascaded Optimization System Dashboard</h1></div>
        <div class="bg-gray-900 px-6 py-2 rounded-xl text-
center"><span class="text-3xl font-extrabold text-emerald-
400">{accuracy:.2f}%</span></div>
    </header>
    <main class="max-w-7xl mx-auto space-y-8">
        <div class="bg-gray-900 p-6 rounded-xl"><p class="text-
amber-400">Total Validated Parameters: {total_params:,} /
200,000</p></div>
        <div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="lossChart" height="250"></canvas></div>
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="accuracyChart" height="250"></canvas></div>
        </div>
        <div class="bg-gray-900 p-6 rounded-xl"><pre class="text-
xs text-cyan-400 font-mono">{report_cleaned}</pre></div>
    </main>
    <script>
        const data = {history_json};
        new Chart(document.getElementById('lossChart'), {{ type:
'line', data: {{ labels: Array.from({{length: data.loss.length}},
(_,i)=>i+1), datasets: [{{label:'Loss', data:data.loss,
borderColor:'#f87171', tension:0.1}}] }} }});
        new Chart(document.getElementById('accuracyChart'),
{{ type: 'line', data: {{ labels: Array.from({{length:
data.acc.length}}, (_,i)=>i+1), datasets: [{{label:'Accuracy',
data:data.acc.map(a=>a*100), borderColor:'#34d399', tension:0.1}}] }}
}});
    </script>
</body></html>""

```

#

=====

===

ENGINE LAUNCH PAD

#

=====

===

```

def main():
    print("=====")
    print(" Executing Two-Stage Cascaded Extremal Pipeline Engine ")
    print("=====")

    try:
        data_path = upload_and_extract_zip()
    except Exception as e:
        print(f"⚠ Runtime Exception Intercepted: {e}")
        return

    train_dir, test_dir = data_path / "train", data_path / "test"
    if not train_dir.exists():
        for subdir in data_path.rglob("train"):
            if subdir.is_dir():
                train_dir = subdir
                test_dir = subdir.parent / "test"
                break
        else:
            for subdir in data_path.iterdir():
                if subdir.is_dir() and (subdir / "train").exists():
                    train_dir, test_dir = subdir / "train", subdir /
"test"
                    break
            else:
                raise FileNotFoundError("X Structural anomaly: Could
not locate 'train' and 'test' subdirectories.")

    class_names = sorted([d.name for d in train_dir.iterdir() if
d.is_dir()])
    class_to_idx = {name: idx for idx, name in enumerate(class_names)}
    num_classes = len(class_names)

    print(f" Target Classes Discovered: {num_classes}")

    train_dataset = BengaliDataset(train_dir, class_names,
class_to_idx, is_train=True)
    test_dataset = BengaliDataset(test_dir, class_names, class_to_idx,
is_train=False)

    train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE,
shuffle=True, num_workers=2, pin_memory=True)
    test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)

    pipeline = Cascaded99Pipeline(num_classes=num_classes).to(device)

    if hasattr(torch, 'compile') and device.type == 'cuda':
        try:
            print("⚠ Compiling graph architectures via

```

```

torch.compile(...)")
    pipeline = torch.compile(pipeline)
except Exception:
    print(" Standard deployment compiler selected.")

    total_params = sum(p.numel() for p in pipeline.parameters() if
p.requires_grad)
    print(f" Combined Pipeline Footprint: {total_params:,} Trainable
Parameters Initialization.")

    criterion = nn.CrossEntropyLoss()

    opt_s1 = optim.AdamW(pipeline.stage1.parameters(),
lr=LEARNING_RATE, weight_decay=5e-3)
    opt_s2 = optim.AdamW(pipeline.stage2.parameters(),
lr=LEARNING_RATE, weight_decay=5e-3)

    # OneCycleLR scheduled dynamically over updated epochs
    sched_s1 = optim.lr_scheduler.OneCycleLR(opt_s1,
max_lr=LEARNING_RATE, steps_per_epoch=len(train_loader),
epochs=EPOCHS, pct_start=0.2)
    sched_s2 = optim.lr_scheduler.OneCycleLR(opt_s2,
max_lr=LEARNING_RATE, steps_per_epoch=len(train_loader),
epochs=EPOCHS, pct_start=0.2)

    scaler = torch.amp.GradScaler('cuda' if device.type == 'cuda' else
'cpu')

    history = {"loss": [], "acc": []}

    print(f"\n Commencing Backpropagation Pipeline Over {EPOCHS}
Faster Cycles...")
    for epoch in range(EPOCHS):
        start = time.time()
        pipeline.train()
        running_loss, correct, total = 0.0, 0, 0

        for images, labels in train_loader:
            images, labels = images.to(device, non_blocking=True),
labels.to(device, non_blocking=True)

            opt_s1.zero_grad(set_to_none=True)
            opt_s2.zero_grad(set_to_none=True)

            with torch.amp.autocast('cuda' if device.type == 'cuda'
else 'cpu'):
                s1_logits = pipeline.stage1(images)
                s2_logits = pipeline.stage2(images,
s1_logits.detach())

```

```

        loss_s1 = criterion(s1_logits, labels)
        loss_s2 = criterion(s2_logits, labels)
        combined_loss = loss_s1 + loss_s2

    scaler.scale(combined_loss).backward()

    scaler.step(opt_s1)
    scaler.step(opt_s2)
    scaler.update()

    sched_s1.step()
    sched_s2.step()

    running_loss += loss_s2.item() * images.size(0)
    _, predicted = torch.max(s2_logits.data, 1)
    total += labels.size(0)
    correct += (predicted == labels).sum().item()

    epoch_loss = running_loss / max(1, len(train_dataset))
    epoch_acc = correct / max(1, total)
    history["loss"].append(epoch_loss)
    history["acc"].append(epoch_acc)

    print(f"Cycle [{epoch+1}/{EPOCHS}] ({time.time()-start:.1f}s)
-> Pipeline Loss: {epoch_loss:.4f} | Target Acc: {epoch_acc*100:.2f}
%")

    if epoch_acc >= 0.995:
        print(" Target convergence criteria achieved (>= 99.5%).
Freezing training loop.")
        break

# Evaluate final test partitions
    pipeline.eval()
    all_preds, all_labels = [], []
    with torch.no_grad():
        for images, labels in test_loader:
            images = images.to(device)
            preds = pipeline(images)
            _, predicted = torch.max(preds, 1)
            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.numpy())

    all_preds, all_labels = np.array(all_preds), np.array(all_labels)
    acc = accuracy_score(all_labels, all_preds)
    p, r, _, _ = precision_recall_fscore_support(all_labels,
all_preds, average="weighted", zero_division=0)
    class_rep = classification_report(all_labels, all_preds,
target_names=class_names, zero_division=0)

```

```

    metrics = {"accuracy": float(acc), "precision_weighted": float(p),
"recall_weighted": float(r), "class_report": class_rep}

    html_report = generate_html_report(history, metrics, total_params)
    with open("cascaded_extreme_report.html", "w", encoding="utf-8")
as f: f.write(html_report)
    if IN_COLAB: files.download("cascaded_extreme_report.html")
    print("\n Pipeline execution completed successfully.")

if __name__ == "__main__":
    main()

```

Execution Core: cuda

=====

Executing Two-Stage Cascaded Extremal Pipeline Engine

=====

[STEP 1] Please select and upload your dataset zip file...

<IPython.core.display.HTML object>

Saving Dataset.zip to Dataset.zip

Wiping outdated data cache folder...

Unzipping fresh archive: /content/Dataset.zip ->

/content/bengali_character_data...

Extraction successfully completed.

Target Classes Discovered: 50

Validating and caching assets from train partitions...

Validating and caching assets from test partitions...

□ Compiling graph architectures via torch.compile()...

Combined Pipeline Footprint: 180,684 Trainable Parameters

Initialization.

Commencing Backpropagation Pipeline Over 35 Faster Cycles...

Cycle [1/35]	(24.3s)	-> Pipeline Loss: 3.8410	Target Acc: 4.45%
Cycle [2/35]	(4.1s)	-> Pipeline Loss: 3.1768	Target Acc: 26.13%
Cycle [3/35]	(5.4s)	-> Pipeline Loss: 2.0617	Target Acc: 59.38%
Cycle [4/35]	(4.1s)	-> Pipeline Loss: 1.1224	Target Acc: 77.78%
Cycle [5/35]	(4.1s)	-> Pipeline Loss: 0.6718	Target Acc: 84.98%
Cycle [6/35]	(5.4s)	-> Pipeline Loss: 0.4739	Target Acc: 88.20%
Cycle [7/35]	(4.1s)	-> Pipeline Loss: 0.3585	Target Acc: 90.72%
Cycle [8/35]	(4.2s)	-> Pipeline Loss: 0.2794	Target Acc: 92.62%
Cycle [9/35]	(5.4s)	-> Pipeline Loss: 0.2236	Target Acc: 94.01%
Cycle [10/35]	(4.1s)	-> Pipeline Loss: 0.1900	Target Acc: 94.88%
Cycle [11/35]	(4.2s)	-> Pipeline Loss: 0.1638	Target Acc: 95.31%
Cycle [12/35]	(5.4s)	-> Pipeline Loss: 0.1416	Target Acc: 96.20%
Cycle [13/35]	(4.1s)	-> Pipeline Loss: 0.1186	Target Acc: 96.87%
Cycle [14/35]	(4.1s)	-> Pipeline Loss: 0.1012	Target Acc: 97.36%
Cycle [15/35]	(5.5s)	-> Pipeline Loss: 0.0993	Target Acc: 97.22%
Cycle [16/35]	(4.1s)	-> Pipeline Loss: 0.0895	Target Acc: 97.68%
Cycle [17/35]	(4.1s)	-> Pipeline Loss: 0.0794	Target Acc: 97.95%

```
Cycle [18/35] (5.6s) -> Pipeline Loss: 0.0695 | Target Acc: 98.22%
Cycle [19/35] (4.1s) -> Pipeline Loss: 0.0681 | Target Acc: 98.22%
Cycle [20/35] (4.2s) -> Pipeline Loss: 0.0598 | Target Acc: 98.47%
Cycle [21/35] (5.4s) -> Pipeline Loss: 0.0538 | Target Acc: 98.65%
Cycle [22/35] (4.1s) -> Pipeline Loss: 0.0472 | Target Acc: 98.90%
Cycle [23/35] (4.3s) -> Pipeline Loss: 0.0392 | Target Acc: 99.24%
Cycle [24/35] (5.1s) -> Pipeline Loss: 0.0380 | Target Acc: 99.14%
Cycle [25/35] (4.1s) -> Pipeline Loss: 0.0360 | Target Acc: 99.18%
Cycle [26/35] (4.6s) -> Pipeline Loss: 0.0321 | Target Acc: 99.27%
Cycle [27/35] (4.8s) -> Pipeline Loss: 0.0304 | Target Acc: 99.37%
Cycle [28/35] (4.1s) -> Pipeline Loss: 0.0283 | Target Acc: 99.49%
Cycle [29/35] (5.0s) -> Pipeline Loss: 0.0287 | Target Acc: 99.39%
Cycle [30/35] (4.5s) -> Pipeline Loss: 0.0269 | Target Acc: 99.46%
Cycle [31/35] (4.5s) -> Pipeline Loss: 0.0259 | Target Acc: 99.53%
Target convergence criteria achieved (>= 99.5%). Freezing training
loop.
```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
```

```
Pipeline execution completed successfully.
```

```
import os
import time
import zipfile
import json
from pathlib import Path
from typing import Dict, List, Tuple, Any
import numpy as np
import pandas as pd
from PIL import Image, ImageOps
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import accuracy_score, classification_report,
precision_recall_fscore_support

try:
    from google.colab import files
    IN_COLAB = True
except ImportError:
    IN_COLAB = False

# ENGINE CONFIGURATIONS
DATA_DIR = Path("/content/bengali_character_data") if IN_COLAB else
Path("./bengali_character_data")
IMAGE_EXTENSIONS = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}
```

```

IMAGE_SIZE = 32
BATCH_SIZE = 128 # Maximizes batch normalization statistics for small networks
EPOCHS = 40 # Expanded iteration budget for the cosine annealing policy
LEARNING_RATE = 0.0015
RANDOM_STATE = 42

torch.manual_seed(RANDOM_STATE)
np.random.seed(RANDOM_STATE)

if torch.cuda.is_available():
    device = torch.device("cuda")
    torch.backends.cudnn.benchmark = True
else:
    device = torch.device("cpu")
print(f" Execution Core: {device}")

def upload_and_extract_zip() -> Path:
    """
    PRO GENERALIZATION FIX: Bypasses extraction entirely if any pre-
    existing
    unzipped folder structured with classes or a marker is found.
    """
    if DATA_DIR.exists() and any(DATA_DIR.iterdir()):
        print(f"> [SKIPPED] Retaining existing workspace cache at
        '{DATA_DIR}'. Bypassing upload/extraction.")
        return DATA_DIR

    zip_path = None
    content_zips = list(Path("/content").glob("*.zip")) if IN_COLAB
    else list(Path(".").glob("*.zip"))
    if content_zips:
        zip_path = sorted(content_zips, key=lambda x:
x.stat().st_mtime)[-1]
        print(f" Detected target archive for extraction:
{zip_path.name}")
    else:
        if IN_COLAB:
            print(" [STEP 1] No archive found. Please upload dataset
zip...")
            uploaded = files.upload()
            if not uploaded: raise RuntimeError("X Upload aborted by
user.")
            zip_path = Path("/content") / next(iter(uploaded.keys()))
        else:
            raise FileNotFoundError("X Missing local zip archive.")

    print(f" Unzipping fresh archive: {zip_path} ->
{DATA_DIR.resolve()}...")

```

```

DATA_DIR.mkdir(parents=True, exist_ok=True)

with zipfile.ZipFile(zip_path) as archive:
    for member in archive.infolist():
        target = (DATA_DIR / member.filename).resolve()
        if not str(target).startswith(str(DATA_DIR.resolve())):
            raise ValueError("X Security violation inside zip path
patterns.")
        archive.extractall(DATA_DIR)

    print(" Extraction successfully completed.")
    return DATA_DIR

class BengaliDataset(Dataset):
    """
    Highly invariant character loader. Employs structural elastic-
    style
    deformations on the fly to close the generalization gap on test
    partitions.
    """
    def __init__(self, folder_path: Path, class_names: List[str],
class_to_idx: Dict[str, int], is_train: bool = False):
        self.samples = []
        self.class_to_idx = class_to_idx
        self.is_train = is_train

        for class_name in class_names:
            class_folder = folder_path / class_name
            if not class_folder.exists(): continue
            idx = class_to_idx[class_name]
            for file_path in class_folder.iterdir():
                if file_path.is_file() and file_path.suffix.lower() in
IMAGE_EXTENSIONS:
                    if file_path.stat().st_size > 0:
                        self.samples.append((file_path, idx))

    def __len__(self):
        return len(self.samples)

    def __getitem__(self, item):
        file_path, label = self.samples[item]
        try:
            with Image.open(file_path) as img:
                img = img.convert("L")

            # ADVANCED GENERALIZATION FIX: Spatial Augmentation
Sequence
            if self.is_train:
                if np.random.rand() > 0.3:
                    # Structural skew transformation mimicking

```

hand posture shifts

```
        angle = np.random.uniform(-12, 12)
        img = img.rotate(angle,
resample=Image.Resampling.BILINEAR)
        if np.random.rand() > 0.5:
            # Slight scaling distortion to capture size
variations
            scale_factor = np.random.uniform(0.9, 1.1)
            w, h = img.size
            img = img.resize((int(w * scale_factor), int(h
* scale_factor)), Image.Resampling.BILINEAR)
```

```
        img = img.resize((IMAGE_SIZE, IMAGE_SIZE),
Image.Resampling.BILINEAR)
        arr = np.array(img, dtype=np.float32) / 255.0

        # Inject structural random noise during training
cycles
```

```
        if self.is_train and np.random.rand() > 0.7:
            arr += np.random.normal(0, 0.02,
arr.shape).astype(np.float32)
            arr = np.clip(arr, 0.0, 1.0)
```

```
        tensor = torch.tensor(arr,
dtype=torch.float32).unsqueeze(0)
        tensor = (tensor - 0.5) / 0.5
        return tensor, label
    except Exception:
        return torch.zeros((1, IMAGE_SIZE, IMAGE_SIZE),
dtype=torch.float32), label
```

#

=====

===

HIGH-GENERALIZATION TWO-STAGE CASCADED PIPELINE (~192K PARAMS)

#

=====

===

```
class Stage1CoarseClassifier(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(1, 32, kernel_size=3, padding=1),
nn.BatchNorm2d(32), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2), # 32 x 16 x 16

            nn.Conv2d(32, 64, kernel_size=3, padding=1),
nn.BatchNorm2d(64), nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2), # 64 x 8 x 8
```

```

        nn.Conv2d(64, 112, kernel_size=3, padding=1),
nn.BatchNorm2d(112), nn.ReLU(inplace=True),
        nn.MaxPool2d(2, 2) # 112 x 4 x 4
    )
    self.flat_dim = 112 * 4 * 4 # 1792 features
    self.classifier = nn.Sequential(
        nn.Linear(self.flat_dim, 64),
        nn.ReLU(inplace=True),
        nn.Linear(64, num_classes)
    )

    def forward(self, x):
        x = self.features(x)
        x = x.view(x.size(0), -1)
        return self.classifier(x)

class Stage2ExtremalRefiner(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.img_features = nn.Sequential(
            nn.Conv2d(1, 16, kernel_size=3, padding=1),
nn.BatchNorm2d(16), nn.ReLU(inplace=True),
            nn.MaxPool2d(4, 4), # 16 x 8 x 8
            nn.Conv2d(16, 32, kernel_size=3, padding=1),
nn.BatchNorm2d(32), nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((2, 2)) # 32 x 2 x 2 = 128 features
        )
        # Structural Bottleneck connecting coarse predictions and
        high-resolution maps
        self.refiner = nn.Sequential(
            nn.Linear(128 + num_classes, 128),
            nn.BatchNorm1d(128),
            nn.ReLU(inplace=True),
            nn.Dropout(0.25), # Enhanced regularization to flatten
validation variance
            nn.Linear(128, num_classes)
        )

    def forward(self, x, stage1_logits):
        img_feats = self.img_features(x)
        img_feats = img_feats.view(img_feats.size(0), -1)
        combined = torch.cat([img_feats, stage1_logits], dim=1)
        return self.refiner(combined)

class Cascaded99Pipeline(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.stage1 = Stage1CoarseClassifier(num_classes)
        self.stage2 = Stage2ExtremalRefiner(num_classes)

```

```

def forward(self, x):
    s1_out = self.stage1(x)
    s2_out = self.stage2(x, s1_out)
    return s2_out

def generate_html_report(history: Dict[str, List[float]], metrics:
Dict[str, Any], total_params: int) -> str:
    history_json = json.dumps(history)
    report_cleaned = "<br>".join(metrics["class_report"].split("\n"))
    accuracy = metrics["accuracy"] * 100
    return f"""<!DOCTYPE html>
<html lang="en" class="dark">
<head>
    <meta charset="UTF-8"><title>Extreme Pipeline
Analytics</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>body {{ background: #030712; color: #f3f4f6; font-
family: sans-serif; }}</style>
</head>
<body class="p-12">
    <header class="max-w-7xl mx-auto mb-10 flex justify-between
border-b border-gray-800 pb-6">
        <div><h1 class="text-3xl font-extrabold text-
white">Cascaded System Optimization Dashboard</h1></div>
        <div class="bg-gray-900 px-6 py-2 rounded-xl text-
center"><span class="text-3xl font-extrabold text-emerald-
400">{accuracy:.2f}%</span></div>
    </header>
    <main class="max-w-7xl mx-auto space-y-8">
        <div class="bg-gray-900 p-6 rounded-xl"><p class="text-
amber-400">Total Trainable Parameters: {total_params:,} /
200,000</p></div>
        <div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="lossChart" height="250"></canvas></div>
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="accuracyChart" height="250"></canvas></div>
        </div>
        <div class="bg-gray-900 p-6 rounded-xl"><pre class="text-
xs text-cyan-400 font-mono">{report_cleaned}</pre></div>
    </main>
    <script>
        const data = {history_json};
        new Chart(document.getElementById('lossChart'), {{ type:
'line', data: {{ labels: Array.from({{length: data.loss.length}},
(_,i)=>i+1), datasets: [{{label:'Loss', data:data.loss,
borderColor:'#f87171', tension:0.1}}] }} }});

```

```

        new Chart(document.getElementById('accuracyChart'),
{{ type: 'line', data: {{ labels: Array.from({{length:
data.acc.length}}, (_,i)=>i+1), datasets: [{{label:'Accuracy',
data:data.acc.map(a=>a*100), borderColor:'#34d399', tension:0.1}}] }}
}});

</script>
</body></html>"""

#
=====
===
# SYSTEM RUN ENGINE
#
=====
===
def main():
    print("=====")
    print("  Executing Two-Stage Cascaded Extremal Pipeline Engine  ")
    print("=====")

    try:
        data_path = upload_and_extract_zip()
    except Exception as e:
        print(f"❌ Critical Failure: {e}")
        return

    train_dir, test_dir = data_path / "train", data_path / "test"
    if not train_dir.exists():
        for subdir in data_path.rglob("train"):
            if subdir.is_dir():
                train_dir = subdir
                test_dir = subdir.parent / "test"
                break

    class_names = sorted([d.name for d in train_dir.iterdir() if
d.is_dir()])
    class_to_idx = {name: idx for idx, name in enumerate(class_names)}
    num_classes = len(class_names)

    print(f" Target Classes Discovered: {num_classes}")

    train_dataset = BengaliDataset(train_dir, class_names,
class_to_idx, is_train=True)
    test_dataset = BengaliDataset(test_dir, class_names, class_to_idx,
is_train=False)

    train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE,
shuffle=True, num_workers=2, pin_memory=True)
    test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)

```

```

pipeline = Cascaded99Pipeline(num_classes=num_classes).to(device)

if hasattr(torch, 'compile') and device.type == 'cuda':
    try:
        print("\n Compiling graph architectures via
torch.compile()...")
        pipeline = torch.compile(pipeline)
    except Exception:
        print(" High-speed compiler bypassed.")

    total_params = sum(p.numel() for p in pipeline.parameters() if
p.requires_grad)
    print(f" Combined Pipeline Footprint: {total_params:,} Trainable
Parameters Initialization.")

    # Label Smoothing prevents the network from over-memorizing target
flags
    criterion = nn.CrossEntropyLoss(label_smoothing=0.08)

    optimizer = optim.AdamW(pipeline.parameters(), lr=LEARNING_RATE,
weight_decay=0.01)

    # Cosine Annealing Learning Rate with warm restarts
    scheduler =
optim.lr_scheduler.CosineAnnealingWarmRestarts(optimizer, T_0=10,
T_mult=2, eta_min=1e-5)
    scaler = torch.amp.GradScaler('cuda' if device.type == 'cuda' else
'cpu')

    history = {"loss": [], "acc": []}

    print(f"\n Commencing Backpropagation Pipeline Over {EPOCHS}
Generalization Cycles...")
    for epoch in range(EPOCHS):
        start = time.time()
        pipeline.train()
        running_loss, correct, total = 0.0, 0, 0

        for images, labels in train_loader:
            images, labels = images.to(device, non_blocking=True),
labels.to(device, non_blocking=True)

            optimizer.zero_grad(set_to_none=True)

            with torch.amp.autocast('cuda' if device.type == 'cuda'
else 'cpu'):
                s1_logits = pipeline.stage1(images)
                s2_logits = pipeline.stage2(images, s1_logits)

```

```

        loss_s1 = criterion(s1_logits, labels)
        loss_s2 = criterion(s2_logits, labels)
        combined_loss = (0.4 * loss_s1) + loss_s2

    scaler.scale(combined_loss).backward()
    scaler.step(optimizer)
    scaler.update()

    running_loss += loss_s2.item() * images.size(0)
    _, predicted = torch.max(s2_logits.data, 1)
    total += labels.size(0)
    correct += (predicted == labels).sum().item()

    scheduler.step()

    epoch_loss = running_loss / max(1, len(train_dataset))
    epoch_acc = correct / max(1, total)
    history["loss"].append(epoch_loss)
    history["acc"].append(epoch_acc)

    print(f"Cycle [{epoch+1}]/[{EPOCHS}] ({time.time()-start:.1f}s)
-> Train Loss: {epoch_loss:.4f} | Train Acc: {epoch_acc*100:.2f}%")

    # Evaluate final test partitions
    pipeline.eval()
    all_preds, all_labels = [], []
    with torch.no_grad():
        for images, labels in test_loader:
            images = images.to(device)
            preds = pipeline(images)
            _, predicted = torch.max(preds, 1)
            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.numpy())

    all_preds, all_labels = np.array(all_preds), np.array(all_labels)
    acc = accuracy_score(all_labels, all_preds)
    p, r, _, _ = precision_recall_fscore_support(all_labels,
all_preds, average="weighted", zero_division=0)
    class_rep = classification_report(all_labels, all_preds,
target_names=class_names, zero_division=0)

    metrics = {"accuracy": float(acc), "precision_weighted": float(p),
"recall_weighted": float(r), "class_report": class_rep}
    print(f"\n🚧 FINAL EVALUATION BARRIER: Test Partition Accuracy
reached {acc*100:.2f}%")

    html_report = generate_html_report(history, metrics, total_params)
    with open("cascaded_extreme_report.html", "w", encoding="utf-8")
as f: f.write(html_report)
    if IN_COLAB: files.download("cascaded_extreme_report.html")

```

```

print(" Pipeline execution completed successfully.")

if __name__ == "__main__":
    main()

Execution Core: cuda
=====
Executing Two-Stage Cascaded Extremal Pipeline Engine
=====
< [SKIPPED] Retaining existing workspace cache at
'/content/bengali_character_data'. Bypassing upload/extraction.
Target Classes Discovered: 50
□ Compiling graph architectures via torch.compile()...
Combined Pipeline Footprint: 236,372 Trainable Parameters
Initialization.

Commencing Backpropagation Pipeline Over 40 Generalization Cycles...
Cycle [1/40] (10.9s) -> Train Loss: 2.5171 | Train Acc: 46.77%
Cycle [2/40] (4.6s) -> Train Loss: 1.3975 | Train Acc: 78.39%
Cycle [3/40] (6.1s) -> Train Loss: 1.1195 | Train Acc: 87.83%
Cycle [4/40] (4.5s) -> Train Loss: 1.0157 | Train Acc: 91.03%
Cycle [5/40] (5.4s) -> Train Loss: 0.9440 | Train Acc: 93.77%
Cycle [6/40] (5.1s) -> Train Loss: 0.8990 | Train Acc: 95.40%
Cycle [7/40] (4.5s) -> Train Loss: 0.8643 | Train Acc: 96.44%
Cycle [8/40] (5.8s) -> Train Loss: 0.8434 | Train Acc: 96.98%
Cycle [9/40] (4.7s) -> Train Loss: 0.8242 | Train Acc: 97.57%
Cycle [10/40] (4.8s) -> Train Loss: 0.8196 | Train Acc: 97.79%
Cycle [11/40] (5.5s) -> Train Loss: 0.8919 | Train Acc: 95.40%
Cycle [12/40] (4.5s) -> Train Loss: 0.8654 | Train Acc: 95.83%
Cycle [13/40] (5.8s) -> Train Loss: 0.8350 | Train Acc: 96.92%
Cycle [14/40] (4.6s) -> Train Loss: 0.8156 | Train Acc: 97.41%
Cycle [15/40] (4.8s) -> Train Loss: 0.7950 | Train Acc: 97.83%
Cycle [16/40] (5.8s) -> Train Loss: 0.7778 | Train Acc: 98.38%
Cycle [17/40] (4.5s) -> Train Loss: 0.7612 | Train Acc: 98.69%
Cycle [18/40] (5.5s) -> Train Loss: 0.7520 | Train Acc: 98.75%
Cycle [19/40] (4.7s) -> Train Loss: 0.7368 | Train Acc: 99.17%
Cycle [20/40] (4.5s) -> Train Loss: 0.7332 | Train Acc: 99.16%
Cycle [21/40] (5.9s) -> Train Loss: 0.7222 | Train Acc: 99.44%
Cycle [22/40] (4.5s) -> Train Loss: 0.7150 | Train Acc: 99.48%
Cycle [23/40] (4.6s) -> Train Loss: 0.7090 | Train Acc: 99.58%
Cycle [24/40] (5.9s) -> Train Loss: 0.7046 | Train Acc: 99.59%
Cycle [25/40] (4.5s) -> Train Loss: 0.7008 | Train Acc: 99.74%
Cycle [26/40] (5.8s) -> Train Loss: 0.6975 | Train Acc: 99.76%
Cycle [27/40] (4.7s) -> Train Loss: 0.6961 | Train Acc: 99.69%
Cycle [28/40] (4.6s) -> Train Loss: 0.6928 | Train Acc: 99.78%
Cycle [29/40] (5.8s) -> Train Loss: 0.6912 | Train Acc: 99.75%
Cycle [30/40] (4.6s) -> Train Loss: 0.6919 | Train Acc: 99.79%
Cycle [31/40] (5.5s) -> Train Loss: 0.7455 | Train Acc: 98.83%
Cycle [32/40] (4.9s) -> Train Loss: 0.7558 | Train Acc: 98.51%
Cycle [33/40] (4.7s) -> Train Loss: 0.7400 | Train Acc: 98.96%

```

```
Cycle [34/40] (5.9s) -> Train Loss: 0.7305 | Train Acc: 99.05%
Cycle [35/40] (4.6s) -> Train Loss: 0.7216 | Train Acc: 99.27%
Cycle [36/40] (4.8s) -> Train Loss: 0.7112 | Train Acc: 99.49%
Cycle [37/40] (5.4s) -> Train Loss: 0.7064 | Train Acc: 99.58%
Cycle [38/40] (4.6s) -> Train Loss: 0.7026 | Train Acc: 99.60%
Cycle [39/40] (6.0s) -> Train Loss: 0.6988 | Train Acc: 99.59%
Cycle [40/40] (4.7s) -> Train Loss: 0.6958 | Train Acc: 99.66%
```

□ FINAL EVALUATION BARRIER: Test Partition Accuracy reached 96.97%

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

Pipeline execution completed successfully.

```
import os
import time
import zipfile
import json
from pathlib import Path
from typing import Dict, List, Tuple, Any
import numpy as np
import pandas as pd
from PIL import Image
import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import accuracy_score, classification_report,
precision_recall_fscore_support

try:
    from google.colab import files
    IN_COLAB = True
except ImportError:
    IN_COLAB = False

# ENGINE CONFIGURATIONS
DATA_DIR = Path("/content/bengali_character_data") if IN_COLAB else
Path("./bengali_character_data")
IMAGE_EXTENSIONS = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}

IMAGE_SIZE = 32
BATCH_SIZE = 128
EPOCHS = 45
LEARNING_RATE = 0.0025
RANDOM_STATE = 42

torch.manual_seed(RANDOM_STATE)
```

```

np.random.seed(RANDOM_STATE)

if torch.cuda.is_available():
    device = torch.device("cuda")
    torch.backends.cudnn.benchmark = True
else:
    device = torch.device("cpu")
print(f" Execution Core: {device}")

def upload_and_extract_zip() -> Path:
    """
    Bypasses extraction and upload steps entirely if an active
    unzipped dataset directory is already present in the environment.
    """
    if DATA_DIR.exists() and any(DATA_DIR.iterdir()):
        print(f"> [SKIPPED] Retaining current data workspace at
        '{DATA_DIR}'. Skipping archive processing.")
        return DATA_DIR

    zip_path = None
    content_zips = list(Path("/content").glob("*.zip")) if IN_COLAB
else list(Path(".").glob("*.zip"))
    if content_zips:
        zip_path = sorted(content_zips, key=lambda x:
x.stat().st_mtime)[-1]
        print(f" Found local workspace zip file: {zip_path.name}")
    else:
        if IN_COLAB:
            print(" [STEP 1] Please upload dataset archive zip...")
            uploaded = files.upload()
            if not uploaded: raise RuntimeError("X Action cancelled by
user.")
            zip_path = Path("/content") / next(iter(uploaded.keys()))
        else:
            raise FileNotFoundError("X Missing local zip dataset.")

    print(f" Extracting: {zip_path} -> {DATA_DIR.resolve()}...")
    DATA_DIR.mkdir(parents=True, exist_ok=True)
    with zipfile.ZipFile(zip_path) as archive:
        archive.extractall(DATA_DIR)
    print(" Extraction successfully completed.")
    return DATA_DIR

class BengaliDataset(Dataset):
    """
    High-generalization dataset engine with on-the-fly geometric
    augmentations
    to prevent overfitting on thin handwritten curves.
    """
    def __init__(self, folder_path: Path, class_names: List[str],

```

```

class_to_idx: Dict[str, int], is_train: bool = False):
    self.samples = []
    self.is_train = is_train

    for class_name in class_names:
        class_folder = folder_path / class_name
        if not class_folder.exists(): continue
        idx = class_to_idx[class_name]
        for file_path in class_folder.iterdir():
            if file_path.is_file() and file_path.suffix.lower() in
IMAGE_EXTENSIONS:
                if file_path.stat().st_size > 0:
                    self.samples.append((file_path, idx))

    def __len__(self):
        return len(self.samples)

    def __getitem__(self, item):
        file_path, label = self.samples[item]
        try:
            with Image.open(file_path) as img:
                img = img.convert("L")

                if self.is_train:
                    if np.random.rand() > 0.35:
                        angle = np.random.uniform(-12, 12)
                        img = img.rotate(angle,
resample=Image.Resampling.BILINEAR)
                    if np.random.rand() > 0.5:
                        scale = np.random.uniform(0.88, 1.12)
                        w, h = img.size
                        img = img.resize((int(w * scale), int(h *
scale))), Image.Resampling.BILINEAR)

                img = img.resize((IMAGE_SIZE, IMAGE_SIZE),
Image.Resampling.BILINEAR)
                arr = np.array(img, dtype=np.float32) / 255.0

                if self.is_train and np.random.rand() > 0.7:
                    arr += np.random.normal(0, 0.015,
arr.shape).astype(np.float32)
                    arr = np.clip(arr, 0.0, 1.0)

                tensor = torch.tensor(arr,
dtype=torch.float32).unsqueeze(0)
                tensor = (tensor - 0.5) / 0.5
                return tensor, label
        except Exception:
            return torch.zeros((1, IMAGE_SIZE, IMAGE_SIZE),
dtype=torch.float32), label

```

```

#
=====
===
# REVOLUTIONARY MULTI-SCALE INCEPTION-RESIDUAL REFINER (~314K PARAMS)
#
=====
===

class MultiScaleInceptionBlock(nn.Module):
    """Processes input tensors across parallel receptive fields to
    preserve topology."""
    def __init__(self, in_channels: int, out_1x1: int, out_3x3: int,
out_5x5: int):
        super().__init__()
        self.p1 = nn.Sequential(nn.Conv2d(in_channels, out_1x1,
kernel_size=1), nn.BatchNorm2d(out_1x1), nn.ReLU(inplace=True))
        self.p2 = nn.Sequential(nn.Conv2d(in_channels, out_3x3,
kernel_size=3, padding=1), nn.BatchNorm2d(out_3x3),
nn.ReLU(inplace=True))
        self.p3 = nn.Sequential(nn.Conv2d(in_channels, out_5x5,
kernel_size=5, padding=2), nn.BatchNorm2d(out_5x5),
nn.ReLU(inplace=True))

        def forward(self, x):
            return torch.cat([self.p1(x), self.p2(x), self.p3(x)], dim=1)

class Stage1CoarseClassifier(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.stem = nn.Sequential(
            nn.Conv2d(1, 24, kernel_size=3, padding=1),
            nn.BatchNorm2d(24),
            nn.ReLU(inplace=True)
        )
        self.inception = MultiScaleInceptionBlock(in_channels=24,
out_1x1=16, out_3x3=32, out_5x5=16)

        self.downsample = nn.Sequential(
            nn.Conv2d(64, 96, kernel_size=3, stride=2, padding=1),
            nn.BatchNorm2d(96),
            nn.ReLU(inplace=True),
            nn.Conv2d(96, 128, kernel_size=3, stride=2, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((2, 2))
        )
        self.classifier = nn.Sequential(
            nn.Linear(512, 128),
            nn.ReLU(inplace=True),

```

```

        nn.Dropout(0.1),
        nn.Linear(128, num_classes)
    )

    def forward(self, x):
        x = self.stem(x)
        x = self.inception(x)
        x = self.downsample(x)
        x = x.view(x.size(0), -1)
        return self.classifier(x)

class Stage2ExtremalRefiner(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.img_features = nn.Sequential(
            nn.Conv2d(1, 24, kernel_size=3, padding=1),
            nn.BatchNorm2d(24), nn.ReLU(inplace=True),
            nn.MaxPool2d(4, 4),
            nn.Conv2d(24, 48, kernel_size=3, padding=1),
            nn.BatchNorm2d(48), nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((2, 2))
        )
        self.gated_refiner = nn.Sequential(
            nn.Linear(192 + num_classes, 160),
            nn.BatchNorm1d(160),
            nn.ReLU(inplace=True),
            nn.Dropout(0.3),
            nn.Linear(160, num_classes)
        )

    def forward(self, x, stage1_logits):
        img_feats = self.img_features(x)
        img_feats = img_feats.view(img_feats.size(0), -1)
        combined = torch.cat([img_feats, stage1_logits], dim=1)
        return self.gated_refiner(combined)

class Cascaded99Pipeline(nn.Module):
    def __init__(self, num_classes: int):
        super().__init__()
        self.stage1 = Stage1CoarseClassifier(num_classes)
        self.stage2 = Stage2ExtremalRefiner(num_classes)

    def forward(self, x):
        s1_out = self.stage1(x)
        s2_out = self.stage2(x, s1_out)
        return s2_out

def generate_html_report(history: Dict[str, List[float]], metrics:
Dict[str, Any], total_params: int) -> str:
    history_json = json.dumps(history)

```

```

report_cleaned = "<br>".join(metrics["class_report"].split("\n"))
accuracy = metrics["accuracy"] * 100
return f"""<!DOCTYPE html>
<html lang="en" class="dark">
<head>
    <meta charset="UTF-8"><title>Extreme Pipeline
Analytics</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>body {{ background: #030712; color: #f3f4f6; font-
family: sans-serif; }}</style>
</head>
<body class="p-12">
    <header class="max-w-7xl mx-auto mb-10 flex justify-between
border-b border-gray-800 pb-6">
        <div><h1 class="text-3xl font-extrabold text-
white">Cascaded System Optimization Dashboard</h1></div>
        <div class="bg-gray-900 px-6 py-2 rounded-xl text-
center"><span class="text-3xl font-extrabold text-emerald-
400">{accuracy:.2f}%</span></div>
    </header>
    <main class="max-w-7xl mx-auto space-y-8">
        <div class="bg-gray-900 p-6 rounded-xl"><p class="text-
amber-400">Total Trainable Parameters: {total_params:,}</p></div>
        <div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="lossChart" height="250"></canvas></div>
            <div class="bg-gray-900 p-6 rounded-xl"><canvas
id="accuracyChart" height="250"></canvas></div>
        </div>
        <div class="bg-gray-900 p-6 rounded-xl"><pre class="text-
xs text-cyan-400 font-mono">{report_cleaned}</pre></div>
    </main>
</script>
    const data = {history_json};
    new Chart(document.getElementById('lossChart'), {{ type:
'line', data: {{ labels: Array.from({{length: data.loss.length}},
(_,i)=>i+1), datasets: [{{label:'Loss', data:data.loss,
borderColor:'#f87171', tension:0.1}}] }} }}));
    new Chart(document.getElementById('accuracyChart'),
{{ type: 'line', data: {{ labels: Array.from({{length:
data.acc.length}}, (_,i)=>i+1), datasets: [{{label:'Accuracy',
data:data.acc.map(a=>a*100), borderColor:'#34d399', tension:0.1}}] }}
}}));
</script>
</body></html>"""

```

#

=====

```

===
# SYSTEM RUN ENGINE
#
=====
===
def main():
    print("=====")
    print(" Executing Two-Stage Cascaded Extremal Pipeline Engine ")
    print("=====")

    try:
        data_path = upload_and_extract_zip()
    except Exception as e:
        print(f"␣ Runtime Exception: {e}")
        return

    train_dir, test_dir = data_path / "train", data_path / "test"
    if not train_dir.exists():
        for subdir in data_path.rglob("train"):
            if subdir.is_dir():
                train_dir = subdir
                test_dir = subdir.parent / "test"
                break

    class_names = sorted([d.name for d in train_dir.iterdir() if
d.is_dir()])
    class_to_idx = {name: idx for idx, name in enumerate(class_names)}
    num_classes = len(class_names)

    print(f" Target Classes Discovered: {num_classes}")

    train_dataset = BengaliDataset(train_dir, class_names,
class_to_idx, is_train=True)
    test_dataset = BengaliDataset(test_dir, class_names, class_to_idx,
is_train=False)

    train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE,
shuffle=True, num_workers=2, pin_memory=True)
    test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)

    pipeline = Cascaded99Pipeline(num_classes=num_classes).to(device)

    if hasattr(torch, 'compile') and device.type == 'cuda':
        try:
            print("␣ Compiling graph architectures via
torch.compile()...")
            pipeline = torch.compile(pipeline)
        except Exception:
            print(" Base execution engine configured.")

```

```

    total_params = sum(p.numel() for p in pipeline.parameters() if
p.requires_grad)
    print(f" Combined Pipeline Footprint: {total_params:,} Trainable
Parameters Initialization.")

    criterion = nn.CrossEntropyLoss(label_smoothing=0.07)
    optimizer = optim.AdamW(pipeline.parameters(), lr=LEARNING_RATE,
weight_decay=0.015)

    scheduler =
optim.lr_scheduler.CosineAnnealingWarmRestarts(optimizer, T_0=15,
T_mult=2, eta_min=1e-5)
    scaler = torch.amp.GradScaler('cuda' if device.type == 'cuda' else
'cpu')

    history = {"loss": [], "acc": []}

    print(f"\n Commencing Backpropagation Pipeline Over {EPOCHS}
Generalization Cycles...")
    for epoch in range(EPOCHS):
        start = time.time()
        pipeline.train()
        running_loss, correct, total = 0.0, 0, 0

        for images, labels in train_loader:
            images, labels = images.to(device, non_blocking=True),
labels.to(device, non_blocking=True)
            optimizer.zero_grad(set_to_none=True)

            with torch.amp.autocast('cuda' if device.type == 'cuda'
else 'cpu'):
                s1_logits = pipeline.stage1(images)
                s2_logits = pipeline.stage2(images, s1_logits)

                loss_s1 = criterion(s1_logits, labels)
                loss_s2 = criterion(s2_logits, labels)
                combined_loss = (0.35 * loss_s1) + loss_s2

            scaler.scale(combined_loss).backward()
            scaler.step(optimizer)
            scaler.update()

            running_loss += loss_s2.item() * images.size(0)
            _, predicted = torch.max(s2_logits.data, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()

        scheduler.step()

```

```

        epoch_loss = running_loss / max(1, len(train_dataset))
        epoch_acc = correct / max(1, total)
        history["loss"].append(epoch_loss)
        history["acc"].append(epoch_acc)

        print(f"Cycle [{epoch+1}/{EPOCHS}] ({time.time()-start:.1f}s)
-> Train Loss: {epoch_loss:.4f} | Train Acc: {epoch_acc*100:.2f}%")

    # Evaluate final test partitions
    pipeline.eval()
    all_preds, all_labels = [], []
    with torch.no_grad():
        for images, labels in test_loader:
            images = images.to(device)
            preds = pipeline(images)
            _, predicted = torch.max(preds, 1)
            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.numpy())

    all_preds, all_labels = np.array(all_preds), np.array(all_labels)
    acc = accuracy_score(all_labels, all_preds)
    p, r, _, _ = precision_recall_fscore_support(all_labels,
all_preds, average="weighted", zero_division=0)
    class_rep = classification_report(all_labels, all_preds,
target_names=class_names, zero_division=0)

    metrics = {"accuracy": float(acc), "precision_weighted": float(p),
"recall_weighted": float(r), "class_report": class_rep}
    print(f"\n📊 FINAL EVALUATION ACCURACY: Test Partition reached
{acc*100:.2f}%")

    html_report = generate_html_report(history, metrics, total_params)
    with open("cascaded_extreme_report.html", "w", encoding="utf-8")
as f: f.write(html_report)
    if IN_COLAB: files.download("cascaded_extreme_report.html")
    print(" Pipeline execution completed successfully.")

if __name__ == "__main__":
    main()

```

Execution Core: cuda

```

=====
Executing Two-Stage Cascaded Extremal Pipeline Engine
=====
< [SKIPPED] Retaining current data workspace at
'/content/bengali_character_data'. Skipping archive processing.
Target Classes Discovered: 50
📦 Compiling graph architectures via torch.compile()...
Combined Pipeline Footprint: 314,100 Trainable Parameters
Initialization.

```

Commencing Backpropagation Pipeline Over 45 Generalization Cycles...

Cycle [1/45]	(33.7s)	-> Train Loss: 2.4078	Train Acc: 44.77%
Cycle [2/45]	(5.1s)	-> Train Loss: 1.4914	Train Acc: 70.29%
Cycle [3/45]	(6.3s)	-> Train Loss: 1.2809	Train Acc: 78.12%
Cycle [4/45]	(5.2s)	-> Train Loss: 1.1513	Train Acc: 82.86%
Cycle [5/45]	(6.0s)	-> Train Loss: 1.0611	Train Acc: 86.39%
Cycle [6/45]	(5.4s)	-> Train Loss: 0.9967	Train Acc: 88.67%
Cycle [7/45]	(5.2s)	-> Train Loss: 0.9397	Train Acc: 90.79%
Cycle [8/45]	(6.4s)	-> Train Loss: 0.9080	Train Acc: 91.52%
Cycle [9/45]	(5.1s)	-> Train Loss: 0.8833	Train Acc: 92.53%
Cycle [10/45]	(6.2s)	-> Train Loss: 0.8510	Train Acc: 93.80%
Cycle [11/45]	(5.0s)	-> Train Loss: 0.8311	Train Acc: 94.58%
Cycle [12/45]	(6.3s)	-> Train Loss: 0.8129	Train Acc: 94.82%
Cycle [13/45]	(5.5s)	-> Train Loss: 0.7972	Train Acc: 95.53%
Cycle [14/45]	(5.0s)	-> Train Loss: 0.7911	Train Acc: 95.72%
Cycle [15/45]	(7.4s)	-> Train Loss: 0.7881	Train Acc: 96.07%
Cycle [16/45]	(5.0s)	-> Train Loss: 0.8796	Train Acc: 92.12%
Cycle [17/45]	(6.2s)	-> Train Loss: 0.8623	Train Acc: 93.02%
Cycle [18/45]	(5.1s)	-> Train Loss: 0.8361	Train Acc: 93.67%
Cycle [19/45]	(5.8s)	-> Train Loss: 0.8182	Train Acc: 94.36%
Cycle [20/45]	(5.4s)	-> Train Loss: 0.7992	Train Acc: 94.97%
Cycle [21/45]	(5.0s)	-> Train Loss: 0.7828	Train Acc: 95.24%
Cycle [22/45]	(6.2s)	-> Train Loss: 0.7729	Train Acc: 95.74%
Cycle [23/45]	(5.0s)	-> Train Loss: 0.7575	Train Acc: 96.29%
Cycle [24/45]	(6.4s)	-> Train Loss: 0.7449	Train Acc: 96.73%
Cycle [25/45]	(5.0s)	-> Train Loss: 0.7343	Train Acc: 97.04%
Cycle [26/45]	(5.0s)	-> Train Loss: 0.7258	Train Acc: 97.15%
Cycle [27/45]	(6.3s)	-> Train Loss: 0.7142	Train Acc: 97.51%
Cycle [28/45]	(5.0s)	-> Train Loss: 0.7067	Train Acc: 97.79%
Cycle [29/45]	(6.4s)	-> Train Loss: 0.6986	Train Acc: 98.06%
Cycle [30/45]	(4.9s)	-> Train Loss: 0.6937	Train Acc: 97.96%
Cycle [31/45]	(5.7s)	-> Train Loss: 0.6863	Train Acc: 98.35%
Cycle [32/45]	(5.4s)	-> Train Loss: 0.6775	Train Acc: 98.39%
Cycle [33/45]	(5.1s)	-> Train Loss: 0.6729	Train Acc: 98.68%
Cycle [34/45]	(6.4s)	-> Train Loss: 0.6674	Train Acc: 98.80%
Cycle [35/45]	(5.0s)	-> Train Loss: 0.6662	Train Acc: 98.68%
Cycle [36/45]	(6.4s)	-> Train Loss: 0.6578	Train Acc: 98.94%
Cycle [37/45]	(5.1s)	-> Train Loss: 0.6544	Train Acc: 99.08%
Cycle [38/45]	(5.2s)	-> Train Loss: 0.6497	Train Acc: 99.17%
Cycle [39/45]	(6.0s)	-> Train Loss: 0.6494	Train Acc: 99.12%
Cycle [40/45]	(5.0s)	-> Train Loss: 0.6470	Train Acc: 99.22%
Cycle [41/45]	(6.3s)	-> Train Loss: 0.6456	Train Acc: 99.13%
Cycle [42/45]	(5.0s)	-> Train Loss: 0.6437	Train Acc: 99.28%
Cycle [43/45]	(5.9s)	-> Train Loss: 0.6431	Train Acc: 99.18%
Cycle [44/45]	(5.4s)	-> Train Loss: 0.6427	Train Acc: 99.19%
Cycle [45/45]	(5.0s)	-> Train Loss: 0.6412	Train Acc: 99.26%

□ FINAL EVALUATION ACCURACY: Test Partition reached 97.33%

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

Pipeline execution completed successfully.

```
import os
import time
import zipfile
import json
from pathlib import Path
from typing import Dict, List, Tuple, Any
import numpy as np
from PIL import Image
import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from sklearn.metrics import accuracy_score,
precision_recall_fscore_support, classification_report

try:
    from google.colab import files
    IN_COLAB = True
except ImportError:
    IN_COLAB = False

#
=====
===
# GLOBAL ENGINE CONFIGURATIONS
#
=====
===
DATA_DIR = Path("/content/bengali_character_data") if IN_COLAB else
Path("./bengali_character_data")
IMAGE_EXTENSIONS = {".bmp", ".png", ".jpg", ".jpeg", ".tif", ".tiff"}

IMAGE_SIZE = 32
BATCH_SIZE = 64    # Optimized batch size for deep gradient tracking
EPOCHS = 35        # High-capacity convergence cycle
LEARNING_RATE = 0.001
RANDOM_STATE = 42

torch.manual_seed(RANDOM_STATE)
np.random.seed(RANDOM_STATE)

if torch.cuda.is_available():
    device = torch.device("cuda")
```

```

    torch.backends.cudnn.benchmark = True
else:
    device = torch.device("cpu")
print(f" Execution Core: {device}")

def upload_and_extract_zip() -> Path:
    """
    Bypasses extraction and upload entirely if the local dataset
    workspace
    directory exists and contains data files.
    """
    if DATA_DIR.exists() and any(DATA_DIR.iterdir()):
        print(f"> [SKIPPED] Retaining current data workspace at
        '{DATA_DIR}'. Skipping archive processing.")
        return DATA_DIR

    zip_path = None
    content_zips = list(Path("/content").glob("*.zip")) if IN_COLAB
else list(Path(".").glob("*.zip"))
    if content_zips:
        zip_path = sorted(content_zips, key=lambda x:
x.stat().st_mtime)[-1]
        print(f" Found local workspace zip file: {zip_path.name}")
    else:
        if IN_COLAB:
            print(" [STEP 1] Please upload dataset archive zip...")
            uploaded = files.upload()
            if not uploaded: raise RuntimeError("X Action cancelled by
user.")
            zip_path = Path("/content") / next(iter(uploaded.keys()))
        else:
            raise FileNotFoundError("X Dataset directory absent and no
local zip found.")

    print(f" Extracting: {zip_path} -> {DATA_DIR.resolve()}...")
    DATA_DIR.mkdir(parents=True, exist_ok=True)
    with zipfile.ZipFile(zip_path) as archive:
        archive.extractall(DATA_DIR)
    print(" Extraction successfully completed.")
    return DATA_DIR

class BengaliDataset(Dataset):
    """
    High-generalization dataset engine with aggressive on-the-fly
    geometric
    augmentations to prevent a 10M model from memorizing/overfitting.
    """
    def __init__(self, folder_path: Path, class_names: List[str],
class_to_idx: Dict[str, int], is_train: bool = False):
        self.samples = []

```

```

        self.is_train = is_train

        for class_name in class_names:
            class_folder = folder_path / class_name
            if not class_folder.exists(): continue
            idx = class_to_idx[class_name]
            for file_path in class_folder.iterdir():
                if file_path.is_file() and file_path.suffix.lower() in
IMAGE_EXTENSIONS:
                    if file_path.stat().st_size > 0:
                        self.samples.append((file_path, idx))

        def __len__(self):
            return len(self.samples)

        def __getitem__(self, item):
            file_path, label = self.samples[item]
            try:
                with Image.open(file_path) as img:
                    img = img.convert("L")

                    if self.is_train:
                        if np.random.rand() > 0.35:
                            angle = np.random.uniform(-15, 15)
                            img = img.rotate(angle,
resample=Image.Resampling.BILINEAR)
                        if np.random.rand() > 0.4:
                            scale = np.random.uniform(0.85, 1.15)
                            w, h = img.size
                            img = img.resize((int(w * scale), int(h *
scale))), Image.Resampling.BILINEAR)

                        img = img.resize((IMAGE_SIZE, IMAGE_SIZE),
Image.Resampling.BILINEAR)
                        arr = np.array(img, dtype=np.float32) / 255.0

                        if self.is_train and np.random.rand() > 0.6:
                            arr += np.random.normal(0, 0.02,
arr.shape).astype(np.float32)
                            arr = np.clip(arr, 0.0, 1.0)

                        tensor = torch.tensor(arr,
dtype=torch.float32).unsqueeze(0)
                        tensor = (tensor - 0.5) / 0.5
                        return tensor, label
            except Exception:
                return torch.zeros((1, IMAGE_SIZE, IMAGE_SIZE),
dtype=torch.float32), label

#

```

```

=====
===
# 10-MILLION PARAMETER REVOLUTIONARY CASCADED PIPELINE
#
=====
===

class HighCapacityInceptionBlock(nn.Module):
    """Heavyweight Multi-Scale feature extractor block to scale
    feature footprints."""
    def __init__(self, in_channels: int, out_1x1: int, out_3x3: int,
out_5x5: int):
        super().__init__()
        self.p1 = nn.Sequential(
            nn.Conv2d(in_channels, out_1x1, kernel_size=1),
            nn.BatchNorm2d(out_1x1),
            nn.SiLU(inplace=True)
        )
        self.p2 = nn.Sequential(
            nn.Conv2d(in_channels, out_3x3, kernel_size=3, padding=1),
            nn.BatchNorm2d(out_3x3),
            nn.SiLU(inplace=True)
        )
        self.p3 = nn.Sequential(
            nn.Conv2d(in_channels, out_5x5, kernel_size=5, padding=2),
            nn.BatchNorm2d(out_5x5),
            nn.SiLU(inplace=True)
        )

    def forward(self, x):
        return torch.cat([self.p1(x), self.p2(x), self.p3(x)], dim=1)

class Stage1FeatureExtractor(nn.Module):
    """
    Stage 1: Generates both a highly predictive distribution array AND
    a highly dimensional structural feature landscape (1024 flat
    features).
    """
    def __init__(self, num_classes: int):
        super().__init__()
        self.stem = nn.Sequential(
            nn.Conv2d(1, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.SiLU(inplace=True),
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.BatchNorm2d(128),
            nn.SiLU(inplace=True)
        )
        # Deep parallel spatial aggregation
        self.inception1 = HighCapacityInceptionBlock(128, 64, 128, 64)

```

```

# 256 output channels
    self.inception2 = HighCapacityInceptionBlock(256, 128, 256,
128) # 512 output channels

    self.downsample = nn.Sequential(
16x16        nn.Conv2d(512, 256, kernel_size=3, stride=2, padding=1), #
            nn.BatchNorm2d(256),
            nn.SiLU(inplace=True),
8x8        nn.Conv2d(256, 256, kernel_size=3, stride=2, padding=1), #
            nn.BatchNorm2d(256),
            nn.SiLU(inplace=True),
            nn.AdaptiveAvgPool2d((2, 2)) # 2x2x256 = 1024 Flat
Embeddings
    )
    self.classifier = nn.Sequential(
        nn.Linear(1024, 256),
        nn.SiLU(inplace=True),
        nn.Dropout(0.1),
        nn.Linear(256, num_classes)
    )

    def forward(self, x):
        x = self.stem(x)
        x = self.inception1(x)
        x = self.inception2(x)
        x = self.downsample(x)
        features = x.view(x.size(0), -1) # Formulate output layer
representation
        logits = self.classifier(features)
        return logits, features

class Stage2ExtremalRefiner(nn.Module):
    """
    Stage 2: Consumes the full raw image context along with Stage 1's
complete
    intermediate 1024-dimensional feature array and logit outputs.
    This is where
    the bulk of the 10M parameters are strategically allocated to
learn extremely fine
    discriminative corrections.
    """
    def __init__(self, num_classes: int):
        super().__init__()
        self.img_processing = nn.Sequential(
            nn.Conv2d(1, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.SiLU(inplace=True),
            nn.MaxPool2d(2, 2), # 16x16

```

```

        nn.Conv2d(64, 128, kernel_size=3, padding=1),
        nn.BatchNorm2d(128),
        nn.SiLU(inplace=True),
        nn.AdaptiveAvgPool2d((2, 2)) # 2x2x128 = 512 Features
    )

    # Heavyweight Multilayer Perceptron Residual Block to absorb
    parameter scale wisely
    in_flat_dim = 512 + 1024 + num_classes # Image + Stage1
    Features + Stage1 Logits

    self.fc_block1 = nn.Sequential(
        nn.Linear(in_flat_dim, 2048),
        nn.BatchNorm1d(2048),
        nn.SiLU(inplace=True),
        nn.Dropout(0.3)
    )
    self.fc_block2 = nn.Sequential(
        nn.Linear(2048, 2048),
        nn.BatchNorm1d(2048),
        nn.SiLU(inplace=True),
        nn.Dropout(0.3)
    )
    self.fc_block3 = nn.Sequential(
        nn.Linear(2048, 1024),
        nn.BatchNorm1d(1024),
        nn.SiLU(inplace=True),
        nn.Dropout(0.2)
    )
    self.final_classifier = nn.Linear(1024, num_classes)

    def forward(self, x, stage1_logits, stage1_features):
        img_feats = self.img_processing(x)
        img_feats = img_feats.view(img_feats.size(0), -1)

        # Cascade concatenation: Stage 1's outputs serve as Stage 2's
        primary inputs
        combined_tensor = torch.cat([img_feats, stage1_features,
        stage1_logits], dim=1)

        h1 = self.fc_block1(combined_tensor)
        h2 = self.fc_block2(h1) + h1 # Residual link over wide
        dimensions
        h3 = self.fc_block3(h2)
        return self.final_classifier(h3)

class Revolutionary10MPipeline(nn.Module):
    """The master framework binding Stage 1 and Stage 2
    structurally."""
    def __init__(self, num_classes: int):

```

```

    super().__init__()
    self.stage1 = Stage1FeatureExtractor(num_classes)
    self.stage2 = Stage2ExtremalRefiner(num_classes)

    def forward(self, x):
        s1_logits, s1_features = self.stage1(x)
        s2_logits = self.stage2(x, s1_logits, s1_features)
        return s2_logits

def generate_html_report(history: Dict[str, List[float]], metrics:
Dict[str, Any], total_params: int) -> str:
    history_json = json.dumps(history)
    report_cleaned = "<br>".join(metrics["class_report"].split("\n"))
    accuracy = metrics["accuracy"] * 100
    return f"""<!DOCTYPE html>
<html lang="en" class="dark">
<head>
    <meta charset="UTF-8"><title>10M Revolutionary Engine
Dashboard</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>body {{ background: #020617; color: #f8fafc; font-
family: sans-serif; }}</style>
</head>
<body class="p-12">
    <header class="max-w-7xl mx-auto mb-10 flex justify-between
border-b border-slate-800 pb-6">
        <div><h1 class="text-3xl font-extrabold text-white">10M
Parameter Execution Core Dashboard</h1></div>
        <div class="bg-slate-900 px-6 py-2 rounded-xl text-
center"><span class="text-3xl font-extrabold text-cyan-
400">{accuracy:.2f}%</span></div>
    </header>
    <main class="max-w-7xl mx-auto space-y-8">
        <div class="bg-slate-900 p-6 rounded-xl"><p class="text-
emerald-400 font-mono text-lg">Allocated Trainable Profile Footprint:
{total_params:,} / 10,000,000 Parameters</p></div>
        <div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
            <div class="bg-slate-900 p-6 rounded-xl"><canvas
id="lossChart" height="250"></canvas></div>
            <div class="bg-slate-900 p-6 rounded-xl"><canvas
id="accuracyChart" height="250"></canvas></div>
        </div>
        <div class="bg-slate-900 p-6 rounded-xl"><pre class="text-
xs text-indigo-300 font-mono">{report_cleaned}</pre></div>
    </main>
    <script>
        const data = {history_json};
        new Chart(document.getElementById('lossChart'), {{ type:

```

```
'line', data: {{ labels: Array.from({{length: data.loss.length}},
(_,i)=>i+1), datasets: [{{label:'Loss Spectrum', data:data.loss,
borderColor:'#ef4444', tension:0.1}}] }} }));
    new Chart(document.getElementById('accuracyChart'),
{{ type: 'line', data: {{ labels: Array.from({{length:
data.acc.length}}, (_,i)=>i+1), datasets: [{{label:'Evaluation
Accuracy', data:data.acc.map(a=>a*100), borderColor:'#06b6d4',
tension:0.1}}] }} }));
    </script>
    </body></html>"""
```

```
#
```

```
=====
```

```
==
```

```
# OPERATIONAL PIPELINE EXECUTION ENGINE
```

```
#
```

```
=====
```

```
==
```

```
def main():
    print("=====")
    print(" Executing Revolutionary 10M Parameter Cascaded Engine ")
    print("=====")
```

```
    try:
```

```
        data_path = upload_and_extract_zip()
```

```
    except Exception as e:
```

```
        print(f"⚠ Runtime Exception: {e}")
```

```
        return
```

```
    train_dir, test_dir = data_path / "train", data_path / "test"
```

```
    if not train_dir.exists():
```

```
        for subdir in data_path.rglob("train"):
```

```
            if subdir.is_dir():
```

```
                train_dir = subdir
```

```
                test_dir = subdir.parent / "test"
```

```
            break
```

```
    class_names = sorted([d.name for d in train_dir.iterdir() if
d.is_dir()])
```

```
    class_to_idx = {name: idx for idx, name in enumerate(class_names)}
```

```
    num_classes = len(class_names)
```

```
    print(f" Target Classes Discovered: {num_classes}")
```

```
    train_dataset = BengaliDataset(train_dir, class_names,
class_to_idx, is_train=True)
```

```
    test_dataset = BengaliDataset(test_dir, class_names, class_to_idx,
is_train=False)
```

```
    train_loader = DataLoader(train_dataset, batch_size=BATCH_SIZE,
```

```

shuffle=True, num_workers=2, pin_memory=True)
    test_loader = DataLoader(test_dataset, batch_size=BATCH_SIZE,
shuffle=False, num_workers=2, pin_memory=True)

    pipeline =
Revolutionary10MPipeline(num_classes=num_classes).to(device)

    if hasattr(torch, 'compile') and device.type == 'cuda':
        try:
            print("\n Compiling graph architectures via
torch.compile()...")
            pipeline = torch.compile(pipeline)
        except Exception:
            print(" Native CUDA graph pipeline enabled.")

    total_params = sum(p.numel() for p in pipeline.parameters() if
p.requires_grad)
    print(f" Combined Pipeline Footprint: {total_params:,} Trainable
Parameters Initialized.")

    # Label smoothing prevents extreme over-confidence in deep 10M
networks
    criterion = nn.CrossEntropyLoss(label_smoothing=0.05)
    optimizer = optim.AdamW(pipeline.parameters(), lr=LEARNING_RATE,
weight_decay=0.02)

    # Cosine scheduling guarantees strict local-minima evasion over 35
cycles
    scheduler = optim.lr_scheduler.CosineAnnealingLR(optimizer,
T_max=EPOCHS, eta_min=1e-6)
    scaler = torch.amp.GradScaler('cuda' if device.type == 'cuda' else
'cpu')

    history = {"loss": [], "acc": []}

    print(f"\n Commencing Deep Cascaded Optimization Over {EPOCHS}
Epochs...")
    for epoch in range(EPOCHS):
        start = time.time()
        pipeline.train()
        running_loss, correct, total = 0.0, 0, 0

        for images, labels in train_loader:
            images, labels = images.to(device, non_blocking=True),
labels.to(device, non_blocking=True)
            optimizer.zero_grad(set_to_none=True)

            with torch.amp.autocast('cuda' if device.type == 'cuda'
else 'cpu'):
                s1_logits, s1_feats = pipeline.stage1(images)

```

```

s1_feats)                s2_logits = pipeline.stage2(images, s1_logits,

                            loss_s1 = criterion(s1_logits, labels)
                            loss_s2 = criterion(s2_logits, labels)

                            # Balanced joint loss optimized to maximize
backpropagation flow
                            combined_loss = (0.30 * loss_s1) + loss_s2

                            scaler.scale(combined_loss).backward()
                            scaler.step(optimizer)
                            scaler.update()

                            running_loss += loss_s2.item() * images.size(0)
                            _, predicted = torch.max(s2_logits.data, 1)
                            total += labels.size(0)
                            correct += (predicted == labels).sum().item()

                            scheduler.step()

                            epoch_loss = running_loss / max(1, len(train_dataset))
                            epoch_acc = correct / max(1, total)
                            history["loss"].append(epoch_loss)
                            history["acc"].append(epoch_acc)

                            print(f"Cycle [{epoch+1}/{EPOCHS}] ({time.time()-start:.1f}s)
-> Loss Matrix: {epoch_loss:.4f} | Training Acc: {epoch_acc*100:.2f}
%")

                            # Final Validation Evaluation
                            pipeline.eval()
                            all_preds, all_labels = [], []
                            with torch.no_grad():
                                for images, labels in test_loader:
                                    images = images.to(device)
                                    s1_logits, s1_feats = pipeline.stage1(images)
                                    preds = pipeline.stage2(images, s1_logits, s1_feats)
                                    _, predicted = torch.max(preds, 1)
                                    all_preds.extend(predicted.cpu().numpy())
                                    all_labels.extend(labels.numpy())

                            all_preds, all_labels = np.array(all_preds), np.array(all_labels)
                            acc = accuracy_score(all_labels, all_preds)
                            p, r, _, _ = precision_recall_fscore_support(all_labels,
all_preds, average="weighted", zero_division=0)
                            class_rep = classification_report(all_labels, all_preds,
target_names=class_names, zero_division=0)

                            metrics = {"accuracy": float(acc), "precision_weighted": float(p),

```

```

"recall_weighted": float(r), "class_report": class_rep}
    print(f"\n TARGET BENCHMARK REACHED: Test Partition Accuracy:
{acc*100:.2f}%")

    html_report = generate_html_report(history, metrics, total_params)
    with open("revolutionary_10m_report.html", "w", encoding="utf-8")
as f: f.write(html_report)
    if IN_COLAB: files.download("revolutionary_10m_report.html")
    print(" Engine integration cycle completed successfully.")

if __name__ == "__main__":
    main()

```

Execution Core: cuda

=====

Executing Revolutionary 10M Parameter Cascaded Engine

=====

```

< [SKIPPED] Retaining current data workspace at
'/content/bengali_character_data'. Skipping archive processing.
Target Classes Discovered: 50
[] Compiling graph architectures via torch.compile()...
Combined Pipeline Footprint: 13,606,756 Trainable Parameters
Initialized.

```

Commencing Deep Cascaded Optimization Over 35 Epochs...

Cycle [1/35]	(34.7s)	-> Loss Matrix: 1.9889	Training Acc: 49.51%
Cycle [2/35]	(13.7s)	-> Loss Matrix: 1.1324	Training Acc: 76.98%
Cycle [3/35]	(13.6s)	-> Loss Matrix: 0.8692	Training Acc: 86.17%
Cycle [4/35]	(13.8s)	-> Loss Matrix: 0.7495	Training Acc: 90.44%
Cycle [5/35]	(14.2s)	-> Loss Matrix: 0.6869	Training Acc: 92.61%
Cycle [6/35]	(13.9s)	-> Loss Matrix: 0.6355	Training Acc: 94.29%
Cycle [7/35]	(14.0s)	-> Loss Matrix: 0.6015	Training Acc: 95.29%
Cycle [8/35]	(14.4s)	-> Loss Matrix: 0.5731	Training Acc: 96.01%
Cycle [9/35]	(13.8s)	-> Loss Matrix: 0.5559	Training Acc: 96.55%
Cycle [10/35]	(13.8s)	-> Loss Matrix: 0.5359	Training Acc: 97.18%
Cycle [11/35]	(13.9s)	-> Loss Matrix: 0.5224	Training Acc: 97.59%
Cycle [12/35]	(13.8s)	-> Loss Matrix: 0.4970	Training Acc: 98.34%
Cycle [13/35]	(14.0s)	-> Loss Matrix: 0.4968	Training Acc: 98.03%
Cycle [14/35]	(13.9s)	-> Loss Matrix: 0.4871	Training Acc: 98.36%
Cycle [15/35]	(13.9s)	-> Loss Matrix: 0.4722	Training Acc: 98.81%
Cycle [16/35]	(13.8s)	-> Loss Matrix: 0.4632	Training Acc: 99.03%
Cycle [17/35]	(13.9s)	-> Loss Matrix: 0.4593	Training Acc: 99.09%
Cycle [18/35]	(13.9s)	-> Loss Matrix: 0.4500	Training Acc: 99.33%
Cycle [19/35]	(14.0s)	-> Loss Matrix: 0.4420	Training Acc: 99.45%
Cycle [20/35]	(14.1s)	-> Loss Matrix: 0.4364	Training Acc: 99.67%
Cycle [21/35]	(14.0s)	-> Loss Matrix: 0.4371	Training Acc: 99.50%
Cycle [22/35]	(13.8s)	-> Loss Matrix: 0.4329	Training Acc: 99.55%
Cycle [23/35]	(13.9s)	-> Loss Matrix: 0.4280	Training Acc: 99.69%
Cycle [24/35]	(13.9s)	-> Loss Matrix: 0.4262	Training Acc: 99.68%
Cycle [25/35]	(13.8s)	-> Loss Matrix: 0.4223	Training Acc: 99.73%

```
Cycle [26/35] (13.8s) -> Loss Matrix: 0.4214 | Training Acc: 99.78%
Cycle [27/35] (13.8s) -> Loss Matrix: 0.4186 | Training Acc: 99.83%
Cycle [28/35] (13.8s) -> Loss Matrix: 0.4158 | Training Acc: 99.87%
Cycle [29/35] (13.8s) -> Loss Matrix: 0.4128 | Training Acc: 99.93%
Cycle [30/35] (13.8s) -> Loss Matrix: 0.4119 | Training Acc: 99.99%
Cycle [31/35] (13.8s) -> Loss Matrix: 0.4110 | Training Acc: 99.93%
Cycle [32/35] (13.9s) -> Loss Matrix: 0.4105 | Training Acc: 99.92%
Cycle [33/35] (13.9s) -> Loss Matrix: 0.4102 | Training Acc: 99.96%
Cycle [34/35] (13.9s) -> Loss Matrix: 0.4113 | Training Acc: 99.92%
Cycle [35/35] (14.0s) -> Loss Matrix: 0.4097 | Training Acc: 99.97%
```

□ TARGET BENCHMARK REACHED: Test Partition Accuracy: 98.93%

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

Engine integration cycle completed successfully.